DE LINEA GEODETICA.

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CONSENSU ET AUCTORITATE

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CAPUT I.

§ 1.

Sphaerois est superficies, rotatione ellipseos circa minorem axem „b“ formata. Quo in motu singula ellipseos peripheriae puncta describunt circulos (parallelos), quorum maximus ab extremis punctis axis majoris „a“ descriptus, aequator appellatur. —

§ 2.

Axis minor ellipseos, circa quem facta est rotatio, nominatur axis sphaeroidis, ejus extrema puncta, poli, omne autem planum, illum axem continens, meridianus.

§ 3.

Angulus „ψ“ is, qui efficitur meridiano, continenti punctum „A“, et (quovis) primo meridiano, appellatur longitudo; angulus „B“ autem, quem normalis in A in sphaeroide erecta cum plano aequatoris effingit, latitudo puncti A. Inde vides, omne punctum sphaeroidis longitudine et latitudine determinari. —

§ 4.

Omnia puncta ejusdem longitudinis ψ, cum in eodem meridiano, et omnia puncta ejusdem latitudinis B in eodem circulo sita sint, illum angulum ψ nominamus longitudinem meridiani, eadem puncta continentis, hunc angulum B latitudinem circuli, eadem puncta continentis. —

§ 5.

Praeter latitudinem B utamur ad situm circuli determinandum alio angulo „l“. Nam, cum, radii loco sphaeroidis, axi „a“ utentes, circa centrum sphaeram describimus, planum circuli sphaeroidis hanc quoque circulo secat. Linearum normalium, in punctis periphe-

riae hujus circuli in sphaera erectarum, unaquaeque effingit cum aequatore angulum, ceteris aequalem, qui „latitudo reducta“ paralleli circuli sphaeroidis appellatur et signo „l“ significatur. —

§ 6.

Circulus enim omnis sphaeroidis, cum et latitudine sua „B“ et latitudine „l“ determinetur, appellatur circulus „B“, sive circulus „l“. Radius circuli „l“ $= a \cos l$.

§ 7.

Altera ex latitudine facile determinatur altera. Siquidem inter angulum l et angulum B ratio intercedit haec:

$$a \operatorname{tg} l = b \operatorname{tg} B$$

vel, cum excentricitas ellipseos meridiani, scilicet $\sqrt{1 - \frac{b^2}{a^2}}$, „$e$“ significatur, ratio haec:

$$\operatorname{tg} l = \sqrt{1 - e^2} \operatorname{tg} B.$$

Quae etsi est simplicissima, latitudines B et l alteram ad alteram reducendi, tamen saepenumero aliis rationibus utemur, quas omnes ad superiorem referri notum est.

Quarum rationum summi momenti sunt hae:

$$\sin l = \frac{b}{a} \frac{\sin B}{\sqrt{1 - e^2 \sin^2 B}}, \qquad \sin B = \frac{\sin l}{\sqrt{1 - e^2 \cos^2 l}},$$

$$\cos l = \frac{\cos B}{\sqrt{1 - e^2 \sin^2 B}}, \qquad \cos B = \frac{b}{a} \frac{\cos l}{\sqrt{1 - e^2 \cos^2 l}},$$

$$\frac{b^2}{a^2} = 1 - e^2 = (1 - e^2 \sin^2 B)(1 - e^2 \cos^2 l).$$

denique posito:

$$\frac{a - b}{a + b} = \frac{1 - \sqrt{1 - e^2}}{1 + \sqrt{1 - e^2}} = \varepsilon,$$

$$B = l + \varepsilon \sin 2l + \frac{\varepsilon^2}{2} \sin 4l + \frac{\varepsilon^3}{3} \sin 6l.$$

§ 8.

Quoniam angulis l et ψ quodque sphaeroidis punctum, ut supra diximus, determinatur, illos coordinatarum loco haberi licet. Ut enim cujuslibet puncti situs in plano duabus coordinatis definitur, et plana curva ratione inter has duas coordinatas intercedente, determinatur, ratio inter l et ψ intercedens exprimat curvam in sphaeroide. Ut porro in discussione planarum curvarum extensio et inclinatio in quoque puncto in axem

aliquem immobilem maxime tractari solent, sic in sphaeroide curvae arcum „s“ (quem, quod ψ longitudo est, extensionem nominare placet); et angulum „T“ (Azimut), quem curva in punctis, in quibus meridianum secat, cum eo efficit, accuratius considerare volumus.

§ 9.

Placet igitur quantitates s et T referre ad quantitates l et ψ. Quare consideremus in aliqua curva sphaeroica duo puncta $A(l, \psi)$ et $B(l + dl, \psi + d\psi)$, quibus nihil propius cogitari potest. Secet meridianus puncti A circulum puncti B in puncto $C(l + dl, \psi)$. Quibus punctis triangulus ABC formatus, cum minimus sit, in plano situs sumi potest. Ejus latera AC et BC, quorum alterum meridianum, alterum circulum tangit, inclinata sunt sub angulo recto.

Itaque:

$$\frac{BC}{AC} = \operatorname{tg} BAC; \qquad \frac{AC}{AB} = \cos BAC; \qquad \frac{BC}{AB} = \sin BAC.$$

$$AB^2 = AC^2 + BC^2.$$

Atqui:

$$BAC = T; \qquad AB = ds,$$

$$BC \text{ (arcus circuli radio „}a\cos l\text{“ descripti)} = a\cos l\, d\psi,$$

$$AC \text{ (arcus ellipseos)} = a\sqrt{1 - e^2\cos^2 l}\, dl.$$

Quibus valoribus superiorum loco positis, fit:

$$ds^2 = a^2(1 - e^2\cos^2 l)\, dl^2 + a^2\cos^2 l\, d\psi^2,$$

$$\operatorname{tg} T = \frac{\cos l}{\sqrt{1 - e^2\cos^2 l}}\frac{d\psi}{dl}; \qquad \cos T = dl\sqrt{\frac{1 - e^2\cos^2 l}{(1 - e^2\cos^2 l)dl^2 + \cos^2 l\, d\psi^2}},$$

$$\sin T = d\psi\sqrt{\frac{\cos^2 l}{(1 - e^2\cos^2 l)dl^2 + \cos^2 l\, d\psi^2}}.$$

§ 10.

In terram spectant e calculis ab ill. Bessel propositis rationes hae:

$$\begin{aligned} a &= 3272077,1399 \text{ tois}; & b &= 3261139,3284 \text{ tois}, \\ \log a &= 6,5148235337; & \log b &= 6,5133693539, \\ \log \tfrac{b}{a} &= 9,9985458202; & \log \tfrac{a}{b} &= 0,0014541798, \\ \varepsilon = \tfrac{a-b}{a+b} &= \tfrac{1-\sqrt{1-e^2}}{1+\sqrt{1-e^2}} = 0,001674184767, & & \\ \log e &= 8,9122052075; & \log \varepsilon &= 7,2238033861. \end{aligned}$$

CAPUT II.

§ 1.

Curva puncto Π in sphaeroide descripta, nulla nisi normali vi movente, nominatur linea geodetica. Aequationem invenimus ejiciendo ex motus corporei puncti aequationibus quantitates t et λ, quarum haec vim normalem, illa tempus proportionale habet.

§ 2.

Aequatione igitur sphaeroidis „$f = 0$“ sumta, aequationes motus in coordinatis rectangulis hae sunt:

$$\frac{d_2x}{dt^2} = \lambda\frac{\partial f}{\partial x},$$
$$\frac{d_2y}{dt^2} = \lambda\frac{\partial f}{\partial y},$$
$$\frac{d_2z}{dt^2} = \lambda\frac{\partial f}{\partial z}.$$

quae si deinceps differentialibus

$$dx \qquad dy \qquad dz$$

multiplicatae adduntur; fit:

$$\frac{dx}{dt}\frac{d_2x}{dt^2} + \frac{dy}{dt}\frac{d_2y}{dt^2} + \frac{dz}{dt}\frac{d_2z}{dt^2} = 0$$

vel integrando:

$$\frac{dx}{dt}\frac{dx}{dt} + \frac{dy}{dt}\frac{dy}{dt} + \frac{dz}{dt}\frac{dz}{dt} = \frac{1}{C^2}$$

sive:

$$dt = C\,ds,$$

unde videmus notum illud, celeritatem puncti Π esse constantem.

§ 3.

Hoc valore in aequationes motus inducto, habemus:

$$\frac{d_2x}{ds^2} = \lambda C^2\frac{\partial f}{\partial x},$$
$$\frac{d_2y}{ds^2} = \lambda C^2\frac{\partial f}{\partial y},$$
$$\frac{d_2z}{ds^2} = \lambda C^2\frac{\partial f}{\partial z}.$$

quibus nihil aliud efficitur, nisi, ut variatio extensionis, qua in constituenda ratio $f = 0$ teneatur, ipsa evanescat.

Qua re aequatio lineae geodeticae est:

$$\delta \int \left\{ \sqrt{dx^2 + dy^2 + dz^2} + \lambda f \right\} = 0$$

vel si pro coordinatis rectangulis „x, y, z“ ellipticis l, ψ utimur

$$\delta \int \sqrt{(1 - e^2 \cos l)\, dl^2 + \cos^2 l\, d\psi^2} = 0.$$

§ 4.

Hic, si usitatas analistorum rationes adhibemus, ad extremum efficitur:

$$\frac{\cos^2 l \, \frac{d\psi}{dl}}{\sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \frac{d\psi}{dl}}} = \text{Const}$$

qua in formula adhibito:

$$\frac{\cos l \, \frac{d\psi}{dl}}{\sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \frac{d\psi}{dl}}} = \sin T$$

conf. Cap. I. § 9.

est

$$\cos l \sin T = \text{Const}$$

et si

$$T = \frac{\pi}{2} \quad \text{sumto} \qquad l = l_0 \quad \text{ponimus,}$$

$$\cos l \sin T = \cos l_0,$$

$$\cos l \cos T = \sqrt{\sin^2 l_0 - \sin^2 l}, \qquad \operatorname{tg} T = \frac{\cos l_0}{\sqrt{\sin^2 l_0 - \sin^2 l}}.$$

§ 5.

Hoc est; linea geodetica circulum aliquem tangens; tangit alium quoque circulum, qui situm illi convenientem habet. Porro, quod semper:

$$l_0 > l$$

est, nunquam fieri potest, ut linea geodetica ultra hos circulos ab aequatore recedat. Propterea lineam geodeticam circulum l_0 tangentem nominamus lineam geodeticam l_0 et punctum illud, quo circulus tangitur, exordium lineae geodeticae, unde deinceps punctorum et extensiones „s“ et longitudines ψ metiemur.

§ 6.

Ad quas e latitudine l determinandas, formulis II ad formulas Cap. I. § 9 propositas relatis, est:

$$a\cos l\,dl\sqrt{1-e^2\cos^2 l} = -\sqrt{\sin^2 l_0 - \sin^2 l}\,ds,$$
$$\cos l_0\,dl\sqrt{1-e^2\cos^2 l} = \mp\sqrt{\sin^2 l_0 - \sin^2 l}\,\cos l\,d\psi,$$

quarum in posteriori signum aut negativum aut positivum ponendum est, prout longitudines, puncto Π ab exordio recedente, aut augentur, aut minuuntur. Integrando fit:

$$s = -a\int_0^l \frac{dl\cos l\sqrt{1-e^2\cos^2 l}}{\sqrt{\sin^2 l_0 - \sin^2 l}},$$
$$\psi = \mp\int_0^l \frac{dl\cos l_0\sqrt{1-e^2\cos^2 l}}{\cos l\sqrt{\sin^2 l_0 - \sin^2 l}}.$$

§ 7.

Definito limite l ita, ut habeat valorem et majorem quam „$-l_0$“ et minorem quam „$+l_0$“ ex his formulis uniter apti valores et extensionis et longitudinis deducuntur. Cui consentaneum est punctum Π in motione circulum l uno tantum in loco secare, quod quia punctus constanti celeritate perpetuo movetur, fieri nequit. Qua de causa formulae a nobis propositae pertinent non ad totam extensionem lineae geodeticae l_0, sed ad eam partem, quae sita est inter exordium et punctum, quo tangitur circulus $-l_0$.

Ut igitur illae transformentur in formulas universi generis, pro latitudine l, quam hactenus per se variabilem posuimus, ejusmodi inducenda est variabilis, quae pro ratione puncti Π ab exordio recedentis, crescat. Cujusmodi est angulus φ, qui definitur aequatione:

$$\sin l_0\cos\varphi = \sin l.$$

§ 8.

Recte angulus φ continet eas, quas quaerimus, qualitates. Est enim ratio inter lineam geodeticam et angulum φ haec, ut in exordio $\varphi = 0$ sit, in sectione aequatoris $\varphi = \frac{\pi}{2}$; in tactione circuli $-l_0$, $\varphi = \pi$, secunda in sectione aequatoris $\varphi = \frac{3}{2}\pi$, secunda in tactione circuli l_0, $\varphi = 2\pi$ etc. Similiter linea procedens, cum tangit utrumque circulum in punctis iis, quae nominare placet puncta conversionis, est $\varphi = m\pi$; cum secat aequatorem, $\varphi = \frac{2m+1}{2}\pi$.

§ 9.

Quoniam cuivis puncto lineae geodeticae l_0 certi quidam valores anguli φ respondent, quantitates

$$l \qquad B \qquad T \qquad s \qquad \psi$$

exprimantur necesse est functionibus anguli φ. Atque l quidem determinatur ex aequatione

$$\sin l = \sin l_0 \cos\varphi$$

vel ex his:

$$\cos l = \sqrt{1-\sin^2 l_0 \cos^2\varphi} = \cos l_0 \sqrt{1+\operatorname{tg}^2 l_0 \sin^2\varphi},$$
$$\operatorname{tg} l = \frac{\operatorname{tg} l_0 \cos\varphi}{\sqrt{1+\operatorname{tg}^2 l_0 \sin^2\varphi}} = \frac{\cos\varphi}{\sqrt{\cot^2 l_0 + \sin^2\varphi}},$$
$$\sqrt{\sin^2 l_0 - \sin^2 l} = \sin l_0 \sin\varphi,$$
$$\sqrt{1-e^2\cos^2 l} = \sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi} = \sqrt{1-e^2\cos^2 l_0 - e^2\sin^2 l_0 \sin^2\varphi}.$$

quibus valoribus in aequationes Cap. I. § 7 et Cap. II. § 4 et § 6 propositas relatis, fit:

$$\sqrt{1-e^2}\ \operatorname{tg} B = \frac{\cos\varphi}{\sqrt{\cot^2 l_0 + \sin^2\varphi}},$$

$$\sin B = \frac{\sin l_0 \cos\varphi}{\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}},$$

$$\cos B = \cos l_0 \sqrt{1-e^2}\sqrt{\frac{1+\operatorname{tg}^2 l_0 \sin^2\varphi}{1-e^2+e^2\sin^2 l_0\cos^2\varphi}},$$

$$\operatorname{tg} T = \frac{\cot l_0}{\sin\varphi}, \qquad \sin T = \frac{1}{\sqrt{1+\operatorname{tg}^2 l_0\sin^2\varphi}}, \qquad \cos T = \frac{\sin\varphi}{\sqrt{\cot^2 l_0 + \sin^2\varphi}},$$

$$s = a\int_0^\varphi d\varphi \sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi} = a\int_0^\varphi d\varphi\sqrt{1-e^2\cos^2 l_0 - e^2\sin^2 l_0\sin^2\varphi},$$

$$\psi = \pm\int_0^\varphi d\varphi\, \frac{\cos l_0}{\sin^2 l_0}\, \frac{\sqrt{1-e^2\cos^2 l_0 - e^2\sin^2 l_0\sin^2\varphi}}{\cot^2 l_0 + \sin^2\varphi}.$$

Postremae duae aequationes, posito:

$$\frac{e\sin l_0}{\sqrt{1-e^2\cos^2 l_0}} = e\sin B_0 = x,$$
$$\sqrt{1-e^2\sin^2 B_0} = \frac{\cos B_0}{\cos l_0} = x',$$

transformantur in:

$$s = \frac{b}{x'}\int_0^\varphi d\varphi\sqrt{1-x^2\sin^2\varphi},$$
$$\psi = \pm\frac{\cot l_0}{\sin B_0}\int_0^\varphi d\varphi\, \frac{\sqrt{1-x^2\sin^2\varphi}}{\cot^2 l_0 + \sin^2\varphi}.$$

CAPUT III.

§ 1.

Nominetur:

$$a\int_0^{\frac{\pi}{2}} d\varphi\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}$$

periodus extensionis et

$$\int_0^{\frac{\pi}{2}} \cos l_0\, d\varphi\,\frac{\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}}{1-\sin^2 l_0\cos^2\varphi}$$

periodus longitudinis lineae geodeticae l_0, quae, cum pendeant tantum ab angulo l_0, significentur $\Sigma(l_0)$ et $\mathcal{A}(l_0)$. Quoniam harum periodorum valores deinceps gravissimi momenti erunt, jam series evolvendae sunt, quibus illi facile quam accuratissime computentur.

§ 2.

Ad inveniendum $\Sigma(l_0)$ sub signo integrationis

$$\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}$$

secundum theorema binomicum evolvimus, ubi, ut breves simus, ponamus:

$$\frac{1}{4}\,\frac{e^2}{1-e^2}=\frac{\varepsilon}{(1-\varepsilon)^2}=o$$

sequitur jam:

$$\begin{aligned}\frac{1}{b}\Sigma(l_0) &= \int_0^{\frac{\pi}{2}} d\varphi\{1+2o\sin^2 l_0\cos^2\varphi-2o^2\sin^4 l_0\cos^4\varphi\\ &\qquad +4o^3\sin^6 l_0\cos^6\varphi-10o^4\sin^8 l_0\cos^8\varphi\ldots\}\\ &= \int_0^{\frac{\pi}{2}} d\varphi+2o\sin^2 l_0\int_0^{\frac{\pi}{2}}\cos^2\varphi\,d\varphi\\ &\quad -2o^2\sin^4 l_0\int_0^{\frac{\pi}{2}}\cos^4\varphi\,d\varphi+4o^3\sin^6 l_0\int_0^{\frac{\pi}{2}}\cos^6\varphi\,d\varphi\\ &\quad -10o^4\sin^8 l_0\int_0^{\frac{\pi}{2}}\cos^8\varphi\,d\varphi\ldots\end{aligned}$$

Singula integralia formam praestant hanc:

$$\int_0^{\frac{\pi}{2}}\cos^{2m}\varphi\,d\varphi$$

valent igitur:

$$\frac{\pi}{2^{2m+1}} \cdot \frac{2m \cdot (2m-1) \cdot (2m-2) \cdots (m+1)}{1 \cdot 2 \cdot 3 \cdots m}$$

Qui valores si ad nostram formulam reducuntur, fit:

$$\frac{2}{b\pi}\Sigma(l_0) = 1 + o\sin^2 l_0 - \frac{3}{4}o^2\sin^4 l_0 + \frac{5}{4}o^3\sin^6 l_0 - \frac{175}{64}o^4\sin^8 l_0 \ldots.$$

§ 3.

Inde vides; cum est,

$$\Sigma(l_0') > \Sigma(l_0'')$$

est:

$$l_0' > l_0''$$

cui consentaneum est l_0 uniter aptam esse functionem periodi extensionis, vel lineam geodeticam periodo extensionis plane definiri.

Inversa igitur serie et posito:

$$\frac{2}{b\pi} \cdot \Sigma(l_0) - 1 = u$$

existit:

$$o\sin^2 l_0 = u + \frac{3}{4}u^2 - \frac{1}{8}u^3 + \frac{5}{32}u^4$$

§ 4.

Ad periodum enucleandam longitudinis computantes

$$\frac{d\Sigma(l_0)}{dl_0}$$

adhibita identitate:

$$\frac{d\Sigma(l_0)}{dl_0} = a\cos l_0 \frac{d\mathcal{A}(l_0)}{dl_0}$$

obtinemus

$$\begin{aligned} a\cos l_0 \frac{d\mathcal{A}(l_0)}{dl_0} &= \frac{b}{2} \cdot \pi\cos l_0\{2o\sin l_0 - 3o^2\sin^3 l_0 + \frac{15}{2}o^3\sin^5 l_0 \\ &\qquad - \frac{175}{8}o^4\sin^7 l_0 \ldots\} \\ \frac{d\mathcal{A}(l_0)}{dl_0} &= \frac{b}{a} \cdot \frac{\pi}{2}\sin l_0\{2o - 3o^2\sin^2 l_0 + \frac{15}{2}o^3\sin^4 l_0 \\ &\qquad - \frac{175}{8}o^4\sin^6 l_0 \ldots\} \\ &= \frac{b\pi}{2a}\sin l_0\{(2 - 3o + \frac{15}{2}o^2 - \frac{175}{8}o^3)o + (3 - 15o + \frac{525}{8}o^2)o^2\cos^2 l_0 \\ &\qquad + (\frac{15}{2} - \frac{525}{8}o)o^3\cos^4 l_0 + \frac{175}{8}o^4\cos^6 l_0 \ldots\} \end{aligned}$$

et integrando intra limites 0 et $\frac{\pi}{2}$

$$\begin{aligned} \frac{\pi}{2} - \mathcal{A}(l_0) = \frac{b}{a}\frac{\pi}{2}\{&(2 - 3o + \frac{15}{2}o^2 - \frac{175}{8}o^3)o\cos l_0 + (1 - 5o + \frac{175}{8}o^2)o^2\cos^3 l_0 \\ &+ (\frac{3}{2} - \frac{105}{8}o)o^3\cos^5 l_0 + \frac{25}{8}o^4\cos^7 l_0, \ldots\}. \end{aligned}$$

Quae in serie constantes coefficientes singularum dignitatum functionis „$\cos l_0$“ ad simpliciores formas referuntur ponendo:

$$p = o - \frac{3}{2}o^2 + \frac{15}{4}o^3 - \frac{175}{8}o^4$$

vel inversa serie

$$o = p + \frac{3}{2}p^2 + \frac{3}{4}p^3 - \frac{5}{16}p^4$$

Qua serie loco o in aequationem reducta, fit:

$$\frac{2\mathcal{A}(l_0)}{\pi} = 1 - \frac{b}{a}\{2p\cos l_0 + (1 - 2p + \frac{25}{8}p^2)p^2\cos^3 l_0 + (\frac{3}{2} - \frac{51}{8}p)p^3\cos^5 l_0 + \frac{25}{8}p^4\cos^7 l_0 \ldots\}.$$

§ 5.

Similiter ut e periodo extensionis definitio lineae geodeticae petitur e periodo longitudinis.

Proinde ex $\mathcal{A}(l_0)$ explicite l_0 computari potest; ponendo

$$a\left\{1 - \frac{2\mathcal{A}(l_0)}{\pi}\right\} = 2bv\sqrt{p} \qquad et \qquad \cos l_0\sqrt{p} = x$$

fit:

$$v = x + x^3\left(\frac{1}{2} - p + \frac{25}{16}p^2\right) + x^5\left(\frac{3}{4} - \frac{51}{16}p\right) + \frac{25}{16}x^7 \ldots$$

et, si eam aequationem invertimus

$$x = v + v^3\left(-\frac{1}{2} + p - \frac{25}{16}p^2\right) + v^5\left(\frac{3}{16}p - \frac{33}{32}p^2\right) - \frac{1}{16}v^7.$$

§ 6.

Numeris primo capite explicatis adhibitis, formae existunt hae:

$$o = 0{,}001\ 6798\ 046\ 65 \qquad p = 0{,}001\ 1755\ 897\ 26$$

$$\Sigma(l_0) = \frac{b\pi}{2} + \frac{b\pi}{2\cdot 10^{12}}\{167\ 9804\ 665\sin^2 l_0 - 211\ 6291\sin^4 l_0 + 5920\sin^6 l_0 - 20\sin^8 l_0 \ldots\}$$

$$\mathcal{A}(l_0) = \frac{\pi}{2} - \frac{b\pi}{2\cdot 10^{12}}\{335\ 1179\ 452\cos l_0 + 282\ 6309\cos^3 l_0 + 7125\cos^5 l_0 + 22\cos^7 l_0 \ldots\}.$$

CAPUT IV.

§ 1.

Integrale

$$\int_{l_1}^{l_2} dl\sqrt{1 - e^2\cos^2 l + \cos^2 l\,\frac{d\psi}{dl}\cdot\frac{d\psi}{dl}}$$

quod supra docuimus, significare extensionem intra puncta $A(l_1, \psi_1)$ et $B(l_2, \psi_2)$ cujusvis lineae in sphaeroide, in certa quadam linea minimum fit.

In quam spectant rationes hae:

$$\delta_1 \int_{l_1}^{l_2} dl \sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}} = 0$$

$$\delta_2 \int_{l_1}^{l_2} dl \sqrt{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}}$$
$$= \int_{l_1}^{l_2} \left(\frac{dl \, \delta\left(\frac{d\psi}{dl}\right) \delta\left(\frac{d\psi}{dl}\right) \cos^2 l [1 - e^2 \cos^2 l]}{\{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}\}^{1\frac{1}{2}}} \right) > 0.$$

Quae linea, quoniam propinquis lineis brevior est, nominetur brevissima.

Atque, ne forte error inducatur hoc nomine, eam, qua omnino nulla brevior cogitari potest, omnium brevissimam dicamus, quae pars brevissimae intelligenda est, dum haec ipsa est pars geodeticae.

§ 2.

Cum integralis:

$$\int_{l_1}^{l_2} \left(\frac{dl \, \delta\left(\frac{d\psi}{dl}\right) \delta\left(\frac{d\psi}{dl}\right) \cos^2 l [1 - e^2 \cos^2 l]}{\{1 - e^2 \cos^2 l + \cos^2 l \, \frac{d\psi}{dl} \cdot \frac{d\psi}{dl}\}^{1\frac{1}{2}}} \right)$$

quodque elementum positivum esse constet, totum integrale positivum sit, necesse est. Cui consentaneum esse videatur, omnem lineam geodeticam vel in tota extensione esse brevissimam.

Quod non ita est, quia Cap. II § 7 demonstravimus, nostrae aequationis vim tantum hactenus stare, quoad circulus „$-l_0$“ tangatur linea geodetica.

§ 3.

Quod igitur ex aequatione integrali, quam supra proposuimus, cognosci non potest, quatenus linea geodetica, tacto etiam circulo „$-l_0$“, brevissima sit; regulae universales statuantur oportet. Quod consequemur inducentes, loco variabilis integrationis l, aliam variabilem ψ integrationis, quam pro ratione recedentis ab exordio puncti II perpetuo crescere, manifestum est.

Proinde quaerendum est, quae sit maxima longitudo ψ, usque ad quam perpetuo

$$\delta_2 \int_{\psi_1}^{\psi_2} d\psi \sqrt{(1 - e^2 \cos^2 l) l'^2 + \cos^2 l}$$

$$\begin{aligned}
&\delta l'\delta l'\cos^2 l(1-e^2\cos^2 l)+2\delta l\delta l'\sin l\cos l\, l'\big(e^2 l'^2[1-e^2\cos^2 l]+[1+e^2\cos^2 l]\big)\\
=\int_{\psi_1}^{\psi_2} d\psi\,&\frac{+\delta l^2\big(e^2 l'^4[\cos 2l-e^2\cos^4 l]+l'^2[2e^2\cos^4 l-2\cos 2l]\big)}{\{(1-e^2\cos^2 l)l'^2+\cos^2 l\}^{1\frac{1}{2}}}\\
=\int_{\psi_1}^{\psi_2} d\psi&\{(11)\delta l'\delta l'+2(01)\delta l\delta l'+(00)\delta l\delta l\}>0
\end{aligned}$$

(cujus integralis' in limitibus $\delta l = 0$ est).

§ 4.

Cujus integralis elementa pro valoribus, variationi δl vindicatis, aut positiva aut negativa fieri possunt. Totum integrale sitne positivum, an negativum, hac re definitur, utrum positivorum an negativorum elementorum summa major sit. Quoniam autem quaeritur, quatenus integrale nostrum, cujusvismodi valores integrationis sunt, positivum sit, omnis quaestio eo referenda est, quatenus integrale inveniatur, quod, cum ipsum 0 sit, illi integrali additum, omnia ejus elementa vi positiva afficiat.

§ 5.

Cujus generis integrale est hoc:

$$\int_{\psi_1}^{\psi_2} d\psi\{\delta l\delta l N\}' = \int_{\psi_1}^{\psi_2} d\psi\{2N\delta l\delta l' + N'\delta l\delta l\}$$

quatenus functio N ita definiri potest, ut summa

$$(11)\delta l'\delta l' + 2[(10)+N]\delta l\delta l' + [(00)+N']\delta l\delta l$$

intra totum spatium, N nunquam infinito, integrum quadratum fiat.

Hac enim conditione

$$\int_{\psi_1}^{\psi_2} d\psi\{2N\delta l\delta l' + N'\delta l\delta l\} = N_2\delta l_2\delta l_2 - N_1\delta l_1\delta l_1 = 0$$

fit, quod variationes in limitibus evanescunt.

§ 6.

Summa:

$$(11)\delta l'\delta l' + 2[(10)+N]\delta l\delta l' + [(00)+N']\delta l\delta l$$

tum existit quadratum, quum

$$(11)[(00)+N'] = [(01)+N]^2$$

vel integrando:

$$-N = (01) + (11)\frac{\alpha\frac{\partial l'}{\partial\psi_1} + \beta\frac{\partial l'}{\partial l_0}}{\alpha\frac{\partial l}{\partial\psi} + \beta\frac{\partial l}{\partial l_0}}$$

qua in aequatione α et β arbitrales constantes sunt.

Integrale

$$\int_{\psi_1}^{\psi_2} d\psi\{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\}$$

tamdiu igitur positivum est, quoad constantes α et β hac ratione possint determinari, ut N in spatio a ψ_1 ad ψ_2 nunquam infinitum evadat.

§ 7.

Quocirca functionem N transformare placet.

Quam ad rem differentiamus aequationem lineae geodeticae:

$$\psi - \psi_1 = \int_{\varphi_1}^{\varphi} \frac{d\varphi \cos l_0 \sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

(in qua angulos φ_1 et φ constat definiri sic:

$$\sin l_0 \cos \varphi_1 = \sin l, \qquad \sin l_0 \cos \varphi = \sin l)$$

secundum ψ_1 et l_0 partialiter, qua in differentiatione ψ constans sumendum est.

Unde sequuntur identitates:

$$-1 = \frac{\cos l_0 \sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi} \frac{d\varphi}{d\psi_1} = -\cot l_0 \frac{\sqrt{1 - e^2 \cos^2 l}}{\cos l \sin \varphi} \frac{dl}{d\psi_1}$$

$$0 = e^2 \sin^2 l_0 \int_{\varphi_1}^{\varphi} d\varphi \frac{\cos^2 \varphi}{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}} + \cot \varphi \sqrt{1 - e^2 \cos^2 l} - \cot \varphi_1 \sqrt{1 - e^2 \cos^2 l_1}$$
$$- \cos l_0 \frac{\sqrt{1 - e^2 \cos^2 l}}{\cos l \sin \varphi} \cdot \frac{dl}{dl_0},$$

ex quibus posito:

$$e^2 \sin^2 l_0 \sin \varphi \int_{\varphi_1}^{\varphi} d\varphi \frac{\cos^2 \varphi}{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}$$
$$+ \cos \varphi \sqrt{1 - e^2 \cos^2 l} - \cot \varphi_1 \sin \varphi \sqrt{1 - e^2 \cos^2 l_1} = u.$$

$$\frac{\cos l_0}{\cos l} \sqrt{1 - e^2 \cos^2 l} = v$$

quotientes differentialium inveniuntur:

$$\frac{dl}{d\psi} = \frac{\sin l_0 \sin \varphi}{v}; \qquad \frac{dl}{dl_0} = \frac{u}{v}$$

quibus valoribus ad N relatis, fit:

$$-N = (01) + (11) \frac{\left(\alpha \frac{\sin l_0 \sin \varphi}{v} + \beta \frac{u}{v}\right)'}{\alpha \frac{\sin l_0 \sin \varphi}{v} + \beta \frac{u}{v}}$$
$$= (01) + \frac{(11) \cdot (1 - \sin^2 l_0 \cos^2 \varphi)}{\cos l_0 \sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}} \cdot \frac{v \frac{d}{d\varphi} \left\{\alpha \frac{\sin l_0 \sin \varphi}{v} + \beta \frac{u}{v}\right\}}{\alpha \sin l_0 \sin \varphi + \beta u}.$$

$$= (10) + \frac{(11)\cos^2 l}{\cos l_0 \sqrt{1 - e^2 \cos^2 l}} \left\{ \frac{\alpha \sin l_0 \cos\varphi + \beta \frac{du}{d\varphi}}{\alpha \sin l_0 \sin\varphi + \beta u} - \frac{\frac{dv}{d\varphi}}{v} \right\}.$$

§ 8.

Quia in dextro latere aequationis quantitates:

$$(01) \quad (11) \quad \cos^2 l \quad \operatorname{tg} l_0 \quad \frac{1}{\sqrt{1 - e^2 \cos^2 l}} \quad \cos\varphi \quad \frac{du}{d\varphi} \quad \frac{dv}{d\varphi} \quad \frac{1}{v}$$

perpetuo finitae manent, functio N hoc tantum modo infinita evadere potest, si:

$$\alpha \sin l_0 \sin\varphi + \beta u = 0$$

fit.

Hinc sequitur, ut integrale

$$\int_{\psi_1}^{\psi_2} d\varphi \{(11)\delta l' \delta l' + 2(01)\delta l \delta l' + (00)\delta l \delta l\}$$

tamdiu positivum sit, quamdiu constantes α et β ita determinari possint, ut

$$\alpha \sin l_0 \sin\varphi + \beta u$$

non evanescat, tamdiu igitur, quoad:

$$\frac{u}{\sin l_0 \sin\varphi}$$

non omnes, inter $-\infty$ et $+\infty$ positos, numeros induat, nam si

$$\frac{u}{\sin l_0 \sin\varphi}$$

aliquem valorem „a“ in hoc spatio non induit, posito:

$$\alpha + \beta a = 0$$

nunquam est:

$$\alpha \sin l_0 \sin\varphi + \beta u = 0.$$

§ 9.

Quod

$$\frac{d}{d\varphi} \left\{ \frac{u}{\sin l_0 \sin\varphi} \right\} = -\frac{1 - \sin^2 l_0 \cos^2\varphi}{\sin l_0 \cos l_0 \sin^2\varphi}$$

semper negativum est et posito $\psi = \psi_1$:

$$\frac{u}{\sin l_0 \sin\varphi} = 0;$$

ψ crescente,

$$\frac{u}{\sin l_0 \sin\varphi}$$

induit deinceps omnes valores a 0 usque ad $-\infty$ et, relapsum a $-\infty$ ad $+\infty$, a ∞ ad 0.

Quo itinere nondum percurso, α et β ita possunt determinari, ut functio N finita maneat; excedente vero

$$\frac{u}{\sin l_0 \sin \varphi}$$

ultra 0, alicubi spatii N infinitum exsistat necesse est, ubi illa norma, sub qua

$$\int_{\psi_1}^{\psi} d\psi\{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\} > 0$$

stare desinit.

Quodsi meminimus,

$$\frac{u}{\sin l_0 \sin \varphi}$$

exstitisse, in aequatione lineae geodeticae ψ secundum constantem l_0 ita partialiter differentiando, ut l constans intelligatur, ei, quod superiori paragrapho proposuimus, respondet illud theorema.

> „Linea geodetica tam diu linea brevissima est, quoad aliam proximam, ab eodem puncto exeuntem, lineam geodeticam secet.“

Punctum hujus sectionis nominatur lineae geodeticae terminus et significatur:

$$O(l, \psi).$$

§ 11.

Revera

$$\int_{\psi_1}^{\psi_2} d\psi\{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\}$$

(posito $\psi = \psi_2$) δl ita determinari potest, ut illud evanescat.

Scilicet ponendo

$$\delta l = \alpha \frac{\partial l}{\partial l_0}$$

integrale nostrum formam induit hanc:

$$\alpha^2 \int_{\psi_1}^{\psi_2} d\psi \left\{(11)\frac{\partial l'}{\partial l_0} \cdot \frac{\partial l'}{\partial l_0} + 2(01)\frac{\partial l}{\partial l_0} \cdot \frac{\partial l'}{\partial l_0} + (00)\frac{\partial l}{\partial l_0} \cdot \frac{\partial l}{\partial l_0}\right\}$$

vel identitatem:

$$\begin{aligned} 0 = & \left((10)\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} + (11)'\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0}\right) - \left((01)\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} + (11)\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0}\right), \\ & = \alpha^2 \int_{\psi_1}^{\psi_2} d\psi \left\{-(01)'\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} - (11)'\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} - 2(01)\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} - (11)\frac{\partial l'}{\partial l_0}\frac{\partial l'}{\partial l_0} - (11)\frac{\partial l}{\partial l_0}\frac{\partial l''}{\partial l_0}\right\} \end{aligned}$$

addendo, hanc:

$$= \alpha^2 \int_{\psi_1}^{\psi_2} d\psi \left\{ (00)\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} - (01)'\frac{\partial l}{\partial l_0}\frac{\partial l}{\partial l_0} - (11)'\frac{\partial l}{\partial l_0}\frac{\partial l'}{\partial l_0} - (11)\frac{\partial l}{\partial l_0}\frac{\partial l''}{\partial l_0} \right\}$$

$$= \alpha^2 \int_{\psi_1}^{\psi_2} d\psi \, \frac{\partial l}{\partial l_0} \left\{ \frac{\partial l}{\partial l_0}\big((00) - (01)'\big) - \left((11)\frac{\partial l'}{\partial l_0}\right)' \right\}$$

$$= \alpha^2 \int_{\psi_1}^{\psi_2} d\psi \, \frac{\partial l}{\partial l_0}\frac{\partial}{\partial l_0}\big((0) - (1)'\big) = 0.$$

Unde efficitur hoc:

> „Linearum geodeticarum, eadem puncta extrema habentium, extensiones differunt quantitatibus minimis tertii gradus.“

§ 12.

Postquam igitur intelleximus, δl convenienter definito, et linea geodetica usque ad terminum O producta,

$$U = \int_{\psi_1}^{\psi_2} d\psi \{(11)\delta l'\delta l' + 2(01)\delta l\delta l' + (00)\delta l\delta l\} = 0$$

fieri; demonstrandum erit, linea longius producta,

$$U < 0$$

existere posse.

Proinde bipartientes integrale nostrum ita, ut altera pars contineatur spatio a ψ_1 ad ψ_2, altera a ψ_2 ad ψ, posito:

$$\psi_1 < \psi_2 < \psi$$

sumimus in prima parte:

$$\delta l = \alpha_1 \frac{dl}{d\psi_1} + \beta_1 \frac{dl}{dl_0},$$

in posteriori:

$$\delta l = \alpha_2 \frac{dl}{d\psi_1} + \beta_2 \frac{dl}{dl_0}$$

quibus in formulis

$$\alpha_1 \qquad \alpha_2 \qquad \beta_1 \qquad \beta_2$$

constantes sunt arbitrales. — Inde fit:

$$U = \begin{cases} \displaystyle\int_{\psi_1}^{\psi_2} d\psi \left\{ (11)\left(\alpha_1\frac{dl'}{d\psi_1} + \beta_1\frac{dl'}{dl_0}\right)^2 + 2(01)\left(\alpha_1\frac{dl'}{d\psi_1} + \beta_1\frac{dl'}{dl_0}\right)\left(\alpha_1\frac{dl}{d\psi_1} + \beta_1\frac{dl}{dl_0}\right) + (00)\left(\alpha_1\frac{dl}{d\psi_1} + \beta_1\frac{dl}{dl_0}\right)^2 \right\} \\ \displaystyle\int_{\psi_2}^{\psi} d\psi \left\{ (11)\left(\alpha_2\frac{dl'}{d\psi_1} + \beta_2\frac{dl'}{dl_0}\right)^2 + 2(01)\left(\alpha_2\frac{dl'}{d\psi_1} + \beta_2\frac{dl'}{dl_0}\right)\left(\alpha_2\frac{dl}{d\psi_1} + \beta_2\frac{dl}{dl_0}\right) + (00)\left(\alpha_2\frac{dl}{d\psi_1} + \beta_2\frac{dl}{dl_0}\right)^2 \right\} \end{cases}$$

et integrando, ut in § 11, et notando functiones

$$\frac{dl}{d\psi_1} \qquad \frac{dl}{dl_0} \qquad \frac{dl'}{d\psi_1} \qquad \frac{dl'}{dl_0} \qquad (01) \quad (11)$$

indicibus iisdem, quibus respondentes ψ:

$$U = \begin{cases} (01)_2 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 \right\}^2 + (11)_2 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ \alpha_1 \left(\frac{dl'}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl'}{dl_0}\right)_2 \right\} \\ - (01)_1 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl}{dl_0}\right)_1 \right\}^2 - (11)_1 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl}{dl_0}\right)_1 \right\} \left\{ \alpha_1 \left(\frac{dl'}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl'}{dl_0}\right)_1 \right\} \\ + (01) \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right) + \beta_2 \left(\frac{dl}{dl_0}\right) \right\}^2 + (11) \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right) + \beta_2 \left(\frac{dl}{dl_0}\right) \right\} \left\{ \alpha_2 \left(\frac{dl'}{d\psi_1}\right) + \beta_2 \left(\frac{dl'}{dl_0}\right) \right\} \\ - (01)_2 \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl}{dl_0}\right)_2 \right\}^2 - (11)_2 \left\{ \alpha_2 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ \alpha_2 \left(\frac{dl'}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl'}{dl_0}\right)_2 \right\} \end{cases}$$

§ 13.

Qua in aequatione constantes

$$\alpha_1, \quad \beta_1, \quad \alpha_2, \quad \beta_2,$$

alternis relationibus inter se ita junctae sunt, ut in limitibus

$$\psi = \psi_1 \quad \text{et} \quad \psi = \psi$$

δl evanescat et intra limites sumto

$$\psi = \psi_2,$$

unum tantum valorem habeat.

Cui rei respondet:

$$\alpha_1 \left(\frac{dl}{d\psi_1}\right)_1 + \beta_1 \left(\frac{dl}{dl_0}\right)_1 = 0 \qquad \alpha_2 \left(\frac{dl}{d\psi_1}\right) + \beta_2 \left(\frac{dl}{dl_0}\right) = 0$$

$$\alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 = \alpha_2 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_2 \left(\frac{dl}{dl_0}\right)_2,$$

vel posito:

$$\alpha_1 = \left(\frac{dl}{dl_0}\right)_1 \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right) \left(\frac{dl}{dl_0}\right)_2 \right\} \omega,$$

$$\beta_1 = - \left(\frac{dl}{d\psi_1}\right)_1 \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right) \left(\frac{dl}{dl_0}\right)_2 \right\} \omega,$$

$$\alpha_2 = - \left(\frac{dl}{dl_0}\right) \left\{ \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right)_2 - \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_1 \right\} \omega,$$

$$\beta_2 = \left(\frac{dl}{d\psi_1}\right) \left\{ \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right)_2 - \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_1 \right\} \omega.$$

Quibus valoribus ad aequationem supra propositam relatis, fit:

$$U = (11)_2 \left\{ \alpha_1 \left(\frac{dl}{d\psi_1}\right)_2 + \beta_1 \left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ (\alpha_1 - \alpha_2) \left(\frac{dl''}{d\psi_1}\right)_2 + (\beta_1 - \beta_2) \left(\frac{dl''}{dl_0}\right)_2 \right\}.$$

$$= (11)_2\omega^2 \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right)\left(\frac{dl}{dl_0}\right)_2 \right\} \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_1 - \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right)_2 \right\}$$
$$\times \left\{ \left(\frac{dl}{d\psi_1}\right)_1 \left(\frac{dl}{dl_0}\right) - \left(\frac{dl}{d\psi_1}\right)\left(\frac{dl}{dl_0}\right)_1 \right\} \left\{ \left(\frac{dl}{d\psi_1}\right)_2 \left(\frac{dl''}{dl_0}\right)_2 - \left(\frac{dl''}{d\psi_1}\right)_2 \left(\frac{dl}{dl_0}\right)_2 \right\}$$

et posito (cap. IV. § 7):

$$\frac{dl}{d\psi_1} = \frac{\sin l_0 \sin\varphi}{v}; \qquad \frac{dl}{dl_0} = \frac{u}{v}$$

$$U = (11)_2\omega^2 \left(\frac{\sin l_0 \sin\varphi_2}{v_2}\frac{u}{v} - \frac{\sin l_0 \sin\varphi}{v}\frac{u_2}{v_2} \right) \left(\frac{\sin l_0 \sin\varphi_2}{v_2}\frac{u_1}{v_1} - \frac{\sin l_0 \sin\varphi_1}{v_1}\frac{u_2}{v_2} \right)$$
$$\times \left(\frac{\sin l_0 \sin\varphi_1}{v_1}\frac{u}{v} - \frac{\sin l_0 \sin\varphi}{v}\frac{u_1}{v_1} \right) \left\{ \left(\frac{\sin l_0 \sin\varphi_2}{v_2} \right) \left(\frac{u}{v}\right)'_2 - \left(\frac{\sin l_0 \sin\varphi}{v} \right)'_2 \left(\frac{u_2}{v_2}\right) \right\}.$$

Quodsi reputamus, posito $\psi = \psi_1$, u evanescere, id est

$$u_1 = 0$$

esse; sequitur:

$$U = (11)_2\omega^2 \sin^4 l_0 \left\{ \frac{u\sin\varphi_2}{vv_2} - \frac{u_2\sin\varphi}{vv_2} \right\} \left\{ -\frac{u_2\sin\varphi_1}{v_1v_2} \right\} \left\{ \frac{u\sin\varphi_1}{vv_1} \right\}$$
$$\times \left\{ \frac{u_2}{v_2} \left(\frac{\sin\varphi}{v}\right)'_2 - \frac{\sin\varphi_2}{v_2}\left(\frac{u}{v}\right)'_2 \right\}$$
$$= (11)_2\omega^2 \frac{uu_2 \sin^4 l_0 \sin\varphi_1 \sin\varphi_2}{v^2v_1^2v_2^2} \{u\sin\varphi_2 - u_2\sin\varphi\} \frac{\sin\varphi_2^2}{v_2^2} \cdot \left\{ \frac{u}{\sin\varphi} \right\}'_2$$
$$= (11)_2\omega^2 \frac{uu_2 \sin^4 l_0 \sin\varphi_1^2 \sin\varphi_2^2}{v^2v_1^2v_2^4} \{u\sin\varphi_2 - u_2\sin\varphi\} \left\{ -\frac{\cos^2 l}{\cos l_0 \sin^2\varphi_2} \right\},$$
$$U = (11)_2 \sin^4 l_0 \frac{\sin^2\varphi_1 \sin\varphi_2^2 \sin^2\varphi \cos^2 l}{\cos l_0} \frac{1}{v^2}\frac{u_2}{v_1^2}\frac{1}{v_2^4} \left(-\frac{u_2}{\sin\varphi_2} \right) \cdot \frac{u}{\sin\varphi} \left\{ \frac{u}{\sin\varphi} - \frac{u_2}{\sin\varphi_2} \right\}.$$

§ 14.

Hactenus cum ψ_2, inter ψ et ψ_1 situm, non accuratius tamen definitum, consideravimus, ψ_2 jam angustioribus coërcendum esse videatur limitibus.

Quoniam enim

$$\frac{u}{\sin l_0 \sin\varphi},$$

ut supra demonstravimus, pro ratione crescentis longitudinis deinceps numeros a 0 ad $-\infty$ induit; ψ_2 tam prope ψ_1 situm sumi potest, ut ψ ad valorem ψ_2 producatur, priusquam

$$\frac{u}{\sin\varphi}$$

ad $-\infty$ adducatur.

Quare quoniam

$$\frac{-u_2}{\sin\varphi_2} > 0$$

est, U aut positivum aut negativum erit, prout:

$$\frac{u}{\sin\varphi}\left\{\frac{u}{\sin\varphi}-\frac{u_2}{\sin\varphi_2}\right\}$$

aut positivum aut negativum est.

Cui convenit,

$$\frac{u}{\sin\varphi}$$

non iterum evanescente, U positivum esse debere, provecto autem ultra 0, U aut positivum aut negativum esse posse.

§ 15.

Unde statuimus theorema:

„Cum punctum aliquod, lineae geodeticae proxime situm, per lineas geodeticas jungitur cum punctis A et B; summa extensionum earum linearum, si $\psi < \psi'$ est, major extensione „$A\ B$“; si $\psi > \psi$ est, aut minor aut major.“

CAPUT V.

§ 1.

Jam vero lineae omnium brevissimae, cujus definitionem initio superioris capitis proposuimus, tractanda est discussio. Quam ob rem omnes lineas geodeticas punctum $A(l,\omega)$ continentes considerabimus. Quarum omnes deinceps aequationes eliciuntur, vindicando in:

$$\psi-\omega=\pm\int_{\varphi_1}^{\varphi}d\varphi\cos l_0\frac{\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}}{1-\sin^2 l_0\cos^2\varphi}$$

(in qua angulos φ et φ_1 formulis

$$\sin l_0\cos\varphi=\sin l \qquad \sin l_0\cos\varphi_1=\sin l_1$$

definiri constat) constanti l_0 omnes numeros a 0 ad $\frac{\pi}{2}$; definito autem l_0, attamen aequatio manet anceps, quum integrale tum positivum, tum negativum sumi possit; ut binae lineae geodeticae eundem tangant circulum.

§ 2.

Quarum linearum genera sunt quatuor. Puncto Π enim in earum qualibet ad propinquum aliquod punctum (l,ψ) provecto, est:

$$\begin{array}{ll} \text{aut } l < l_1, & \psi > \omega, \\ \text{aut } l > l_1, & \psi > \omega, \\ \text{aut } l < l_1, & \psi < \omega, \\ \text{aut } l > l_1, & \psi < \omega. \end{array}$$

Aequationes linearum primi et quarti generis formam praestant hanc:

$$\psi - \omega = \int_{\varphi_1}^{\varphi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

secundi et tertii hanc:

$$\omega - \psi = \int_{\varphi_1}^{\varphi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}.$$

§ 3.

Omnium autem linearum geodeticarum ratio est haec, ut unaquaeque omnia puncta zonae, sphaeroidis circulis l_0 et $-l_0$ inclusae contineat.

ARGUMENTATIO.

Demonstrandum est, quamvis lineam geodeticam quemlibet circulum zonae deinceps omnibus punctis secare; longitudines igitur eorum, omnes deinceps numeros, intra 0 et 2Π sitos, induere.

Quae sectionis longitudines petuntur ex aequatione:

$$\psi - \omega = \pm \int_{\varphi_1}^{\varphi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

in qua φ_1 constans est, valores autem anguli φ deducuntur ex aequatione:

$$\sin l = \sin l_0 \cos \varphi.$$

Posito in primo sectione:

$$\varphi = \varphi_2 \qquad \psi = \psi_2,$$

aliis in sectionibus deinceps fiet: angulus

$$\varphi = \varphi_2 + 2\pi \qquad \varphi_2 + 4\pi$$

$$\psi = \omega + \int_{\varphi_1}^{\varphi_2 + 2\pi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

$$\omega + \int_{\varphi_1}^{\varphi_2 + 4\pi} \cos l_0 \, d\varphi \, \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0 \cos^2 \varphi}}{1 - \sin^2 l_0 \cos^2 \varphi}$$

vel $\omega + 4\mathcal{A}(l_0)$ $\qquad$ $\omega + 8\mathcal{A}(l_0)$

vel referendo ad universum genus:

$$\omega + 4n\mathcal{A}(l_0) - 2h\pi$$

qua in formula n deinceps omnium integrorum numerorum valores induit, et h eum numerum integrum repraesentat, qui est in:

$$\frac{\psi_2 + 4n\mathcal{A}(l_0)}{2\pi}.$$

Quod igitur

$$\frac{\mathcal{A}(l_0)}{2\pi}$$

irrationale est (quippe contrarium, quod rarissimo fit, omittendum est); ψ omnes deinceps valores, intra 0 et 2π sitos, assumit; id, quod erat demonstrandum.

§ 4.

Jam vero, priusquam quaerere instituimus, quae sit brevissima omnium linea intra duo puncta $A(l\psi)$ et $B(l_1\psi_1)$ in sphaeroide; nonnulla eam rem spectantia considerare placeat.

Illa in quaestione l majus esse l_1 ponamus. E theoremate superiori cum jam dilucide appareat, inter lineas a puncto A exeuntes omnes punctum B continere; quaerendum est accuratius, qualis ceteris brevior existat. Ubi haec quidem res in promptu est, lineam, quae ceteris sit brevior, genere noto, definiri posse.

ARGUMENTATIO.

Nam si alicujus lineae geodeticae cursum a puncto A atque multarum aliarum linearum geodeticarum prope hanc currentium et minus in meridianum puncti A inclinatarum intuemur, lineam geodeticam a linea aliqua eo prius secari, invenimus, quo minor sit inclinatio in ipsam. Qua e re sequitur, ut omnes lineae geodeticae ejusdem generis hanc post terminum secent. Quum vero quaevis linea geodetica, ut supra demonstratum est, usque ad terminum brevissima sit; eo magis, lineam aliam geodeticam secans, jam longe linea esse brevissima desierit, necesse est.

§ 5.

Priusquam autem ad genus determinandum propius accedamus, duo theoremata proponamus haec.

I. „Linea geodetica l_0 I, si a puncto A exorsa, nondum assecuta punctum B, secat meridianum puncti A (e. g. in C); non est linea omnium brevissima inter A et B.“

ARGUMENTATIO.

Ea linea geodetica II, cujus Azimut negativo Azimut lineae geodeticae I in A aequale est, propter symmetriam aeque ac linea I et tangit circulum l_0 et continet

punctum C. Cujus utriusque lineae extensio intra A et C cum eadem sit; ab A ad B existunt duo aequa itinera, quorum alterum ipsa est linea geodetica I, alterum duabus partibus conficitur, scilicet linea geodetica II intra A et C et linea geodetica I intra C et B.

Quodsi superius iter intra A et B linea omnium brevissima esset, nimirum etiam posterius linea esset omnium brevissima, quod fieri nequit, quia posterius iter in puncto C non est linea geodetica.

II. „Ea linea geodetica l_0 primi aut tertii generis, quae ducta ab A ad B circulum $-l_0$ bis secat (altero quidem loco in C), non est linea omnium brevissima intra A et B.“

Cujus rei demonstrandae ratio priori similis est. Namque puncto II ab A in altera linea geodetica, quae tangit circulum l_0, promota ad C, tum in priori a C ad B; hoc totum iter extensioni AB lineae geodeticae I aequum est.

Est igitur, quod posuimus, superiori argumentandi ratione demonstratum.

§ 6.

Linea, tangens circulum l_1 in puncto A, dividit zonam, circulis l_1 et $-l_1$ inclusam, in duas partes, quarum alteram nominare placet „J“, intra lineam geodeticam, alteram „E“, extra lineam geodeticam sitam.

Quare, vi, quae in theorematis modo propositis inest, percognita, perspicuum est; lineam omnium brevissimam intra A et $B(l\psi)$, B sito in parte „J“, primi aut tertii generis esse neque habere intra A et B punctum conversionis; B sito in parte „E“, secundi aut quarti generis esse et habere intra A et B punctum conversionis; si

$$\psi > \omega$$

primi aut secundi generis esse, si

$$\psi < \omega$$

tertii aut quarti generis esse.

Quaecunque igitur zonae pars punctum B continet, quo in genere linearum geodeticarum ponenda sit linea omnium brevissima intra A et B, definiri potest; ac quidem ea linea geodetica, quae omnium brevissima est, circulum $-l_0$ non secat.

§ 7.

Quae eo spectant, ut, si linea geodetica pars est extensionis periodi, ceteris lineis geodeticis intra A et B sit brevior; vel „Linea geodetica l_0 hac conditione omnium intra A et B brevissima est linea, si angulorum φ, ad haec puncta pertinentium, differentia non excedit ultra 180°.“

CAPUT VI.

§ 1.

E formulis capitis II ab ill. Legendre propositis gravissima momenta lineae geodeticae deduci posse, vidimus.

Ad computandas autem longitudines et extensiones ex l, longe aptiores sunt formulae ab ill. Jacobi evolutae in libello:

„Solution nouvelle d'un problème de Géodésie“

transeundo ab integralibus ellipticis ad functiones ellipticas.

§ 2.

Quem sequentes loco anguli φ per se variabilis inducimus variabilem:

$$u = \int_0^{\varphi} \frac{d\varphi}{\sqrt{1 - x^2 \sin^2 \varphi}}$$

cujus functionibus omnes quantitates exprimamus. Proinde ad computandum φ formulis, inversa aequatione integrali nascentibus, utimur; ad reliquas vero computandas quantitates, referri necesse est hos valores ad aequationes capitis II.

Inde fit:

$$\varphi = \operatorname{am} u, \qquad \sin\varphi = Su, \qquad \cos\varphi = Cu, \qquad \sqrt{1 - x^2 \sin^2 \varphi} = \Delta u,$$

$$\sin l = \sin l_0 \, Cu, \qquad \cos l = \cos l_0 \sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u},$$

$$\operatorname{tg} l = \operatorname{tg} l_0 \frac{Cu}{\sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u}},$$

$$\sqrt{1 - e^2 \cos^2 l} = \frac{b}{a} \cdot \frac{\Delta u}{x'} = \sqrt{1 - e^2 \cos^2 l_0} \, \Delta(u),$$

$$\operatorname{tg} B = \operatorname{tg} B_0 \frac{Cu}{\sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u}},$$

$$\sin B = \sin B_0 \frac{Cu}{\Delta u}, \qquad \cos B = \cos B_0 \frac{\sqrt{1 + \operatorname{tg}^2 l_0 \, S^2 u}}{\Delta u},$$

$$\sqrt{1 - e^2 \sin^2 B} = \frac{\cos B}{\cos l} = \frac{x'}{\Delta u}.$$

$$\operatorname{tg} T = \frac{\cot l_0}{Su}, \qquad \sin T = \frac{1}{\sqrt{1+\operatorname{tg}^2 l_0\, S^2 u}}, \qquad \cos T = \frac{Su}{\sqrt{\cot^2 l_0 + S^2 u}},$$

$$s = \frac{b}{x'}\int_0^u \Delta^2 u\, du,$$

$$\psi = \frac{\cot l_0}{\sin B_0}\int_0^u du\, \frac{\Delta^2 u}{\cot^2 l_0 + S^2 u}.$$

§ 3.

Hac ratione functiones trigonometricae angulorum:

$$B \qquad l \qquad T$$

repraesentantur functionibus ellipticis quantitatis u, cujus Θ functionibus jam extensionem et longitudinem determinare placet.

Quam ob rem formula quantitatis s transformetur in hanc:

$$\frac{x's}{b} = u - \int_0^u x^2 S^2 u\, du$$

in qua, formula adhibita, fund. pag. 133 proposita; et posito:

$$\int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1-x^2\sin^2\varphi}} = K, \qquad \int_0^{\frac{\pi}{2}} d\varphi\sqrt{1-x^2\sin^2\varphi} = E$$

fit:

$$\frac{2Kx'}{b\pi}\cdot s = \frac{2K}{\pi}u + \frac{2K}{\pi}Zu - u\left(\frac{2K}{\pi} - \frac{2E}{\pi}\right) = \frac{2E}{\pi}u + \frac{2K}{\pi}Zu$$

$$s = \frac{b}{x'}\cdot\frac{E}{K}u + \frac{b}{x'}Zu$$

et, si inducitur loco constantis E, constans $\Sigma(l_0)$, in quam s, posito $u = K$ transit; id est, si ponimus

$$\frac{b}{x'}E. = \Sigma(l_0)$$

$$s - \frac{u}{K}\Sigma(l_0) = \frac{b}{x'}Zu.$$

§ 4.

Difficilius est referre longitudinem ad Θ functiones. Quam ad rem, qua usus est ratione in libello suo

Solution nouvelle d'un problème de Géodésie

vir ill. Jacobi, ea paullo discrepat ab illa, quam proposuit in fundamentis.

Quamquam haec tam exigua est discrepantia, ut, superiorem in disquisitione nostra sequentes, nihil artificiosi inducamus. Proinde nova utimur constanti v, definita formula:

$$v = \int_0^{\frac{\pi}{2}-l} \frac{d\varphi}{\sqrt{1-x'^2\sin^2\varphi}}$$

vel his formulis:

$$S(v,x') = \cos l_0; \qquad C(v,x') = \sin l_0; \qquad \Delta(v,x') = \sin B_0.$$

quibus valoribus in aequationes nostras illatis et adhibitis formulis fund. pag. 34 propositis, fit:

$$\begin{aligned}
\psi &= \frac{S(v,x')}{C(v,x')\Delta(v,x')}\int_0^u \frac{\Delta^2 u\,du}{S^2u + \frac{S^2(v,x')}{C^2(v,x')}} \\
&= -i\frac{Siv\,Civ}{\Delta iv}\int_0^u \frac{\Delta^2 u\,du}{S^2u - S^2iv} \\
&= i\frac{\Delta(iv-iK')S(iv-iK')}{C(iv-iK')}\int_0^u \frac{\Delta^2 u\,du}{1-x^2S^2u\,S^2(iv-iK')} \\
&= iu\frac{\Delta(iv-iK')S(iv-iK')}{C(iv-iK')} - iuZ(iv-iK') - \frac{i}{2}\log\frac{\Theta(u-iv+iK')}{\Theta(u+iv-iK')} \qquad \text{cf. fund. p. 146.} \\
&= iu\frac{Civ}{Siv\,\Delta iv} - iu\left(\frac{i\pi}{2K} + \frac{Civ\,\Delta iv}{Siv} + Ziv\right) - \frac{i}{2}\log\frac{-i.e.^{\frac{\pi(K'-2iu-2v)}{4K}}H(u-iv)}{i.e.^{\frac{\pi(K'+2iu-2v)}{4K}}H(u+iv)}
\end{aligned}$$

cf. p. 163; cf. f. p. 164.

$$\begin{aligned}
&= iux^2\frac{Siv\,Civ}{\Delta iv} + \frac{\pi u}{2K} - iuZiv - \frac{i}{2}\left\{\pi\frac{K'-2iu-2v}{4K} - \pi\frac{K'+2iu-2v}{4K}\right\} \\
&\quad - \frac{i}{2}\log\frac{H(iv-u)}{H(iv+u)} = -iuZ(iv+K) + \frac{i}{2}\log\frac{H(iv+u)}{H(iv-u)}.
\end{aligned}$$

Posito: $u = K$ fit $\psi = \mathcal{A}(l_0)$; ergo

$$\mathcal{A}(l_0) = -iKZ(iv+K) + \frac{\pi}{2}$$

et referendo ad superiorem formam:

$$\psi = \frac{u}{K}\left\{\mathcal{A}(l_0) - \frac{\pi}{2}\right\} + \frac{i}{2}\log\frac{H(iv+u)}{H(iv-u)} = \frac{u\mathcal{A}(l_0)}{K} + \frac{i}{2}\log\frac{\Theta(iv-iK'+u)}{\Theta(iv-iK'-u)}.$$

Quam aequationem etsi longe elegantiorem formam ill. Jacobi ostendit induere posse, nos tamen ad rem nostram in illa stare possumus.

§ 5.

Quod quia Θ functiones in series praecipiter convergentes explicantur, ideoque facillime computantur, jam in re erit, etiam functiones trigonometricas angulorum

$$B \qquad l \qquad T$$

referre ad Θ functiones.

Formularum ea ratione evolutarum gravissimae hae sunt:

$$\sin l = \sin l_0 \sqrt{\frac{x'}{x}}\,\frac{H(u+K)}{\Theta(u)},$$
$$\sqrt{x}\sin B = \sin B_0 \frac{H(u+K)}{\Theta(u+K)},$$
$$\frac{\cos B}{\cos l} = \sqrt{x'}\frac{\Theta u}{\Theta(u+K)},$$
$$\operatorname{tg} T = \cot l_0 \sqrt{x'}\,\frac{\Theta u}{H(u)}.$$

CAPUT VII.

§ 1.

Ut vero quantitates:

$$l \qquad s \qquad \psi$$

ex aequationibus capitis superioris ad calculum vocentur, formulas statuere necesse est ad constantes:

$$l_0 \qquad B_0 \qquad \Sigma(l_0) \qquad \mathcal{A}(l_0) \qquad x \qquad x' \qquad K \qquad E \qquad q \qquad v$$

omnes ex una computandas.

§ 2.

Plerumque circulus, qui curva tangitur, definitus est; rarissimo ab hac ratione deceditur. Ubi altera e duabus periodis nota est; (e. g. cum quaeritur: „Quemnam circulum tangit linea omnium brevissima duobus aequatoris punctis definita?“) tum ad computandum l_0 tantummodo seriebus Capit. III utemur.

§ 3.

Perinde, l_0 nunc notum fingentes, determinare possumus quantitates:

$$B_0 \qquad \Sigma(l_0) \qquad \mathcal{A}(l_0) \qquad x \qquad x'$$

ex formulis:

$$\begin{aligned}
\sqrt{1-e^2}\ \mathrm{tg}\, B_0 &= \mathrm{tg}\, l_0, && \text{Cap. I § 7}\\
\tfrac{2}{\pi}\tfrac{\Sigma(l_0)}{b} &= 1 + o\sin^2 l_0 - \tfrac{3}{4}o^2\sin^4 l_0 + \tfrac{5}{4}o^3\sin^6 l_0 \ldots, && \text{Cap. III § 2}\\
\tfrac{a}{b}\left\{1 - \tfrac{2}{\pi}\mathcal{A}(l_0)\right\} &= 2p\cos l_0 + \left\{1 - 2p + \tfrac{25}{8}p^2\right\} p^2\cos^3 l_0 \ldots, && \text{Cap. III § 4}\\
x &= e\sin B_0, && \text{Cap. II § 9}\\
x' &= \tfrac{\cos B_0}{\cos l_0}.
\end{aligned}$$

§ 4.

Restat, ut computentur constantes:

$$q \qquad x \qquad E \qquad v.$$

Quam ad rem formulam adhibere placet hanc:

$$\sqrt{x'} = \frac{1 - 2q + 2q^4 \cdots}{1 + 2q + 2q^4 \cdots} \qquad \text{fund. pag. 184.}$$

Quum q quantitas minima sit, quod quidem ad terram attinet, nunquam numerum „0,0004“ excedens, q^4 omnino negligi licet:

Est igitur:

$$\sqrt{x'} = \frac{1-2q}{1+2q}$$

vel:

$$2q = \frac{1-\sqrt{x'}}{1+\sqrt{x'}} = \frac{\sqrt{\cos l_0} - \sqrt{\cos B_0}}{\sqrt{\cos l_0} + \sqrt{\cos B_0}},$$

unde q facile computari potest.

Qua in re, cum interdum x' non accurate definitum sit, fieri posse concedimus, ut q non extremis numeris computari possit. Quod ne fiat, explicanda est alia formula.

Est:

$$\sqrt{x} = 2\sqrt[4]{q}\,\frac{1+q^2}{1+2q} \qquad \text{fund. pag. 184}$$

$$q = \frac{x^2}{16}\left\{\frac{1+2q}{1+q^2}\right\}^4$$

$$\log q = \log\frac{x^2}{16} - 4\log(1+q^2) + 4\log(1+2q) = \log\frac{x^2}{16} + 8q - 12q^2 + \frac{8}{3}q^3.$$

Sin q accuratius computandum est, computetur oportet quantitas q_1 aliqua, propius ad q accedens, e formula

$$\log q_1 = \log\frac{x^2}{16} + \frac{x^2}{2}$$

quo facto q definitur ex aequatione

$$\log q = \log\frac{x^2}{16} + 8q_1 - 12q_1^2 + \frac{8}{3}q_1^3.$$

§ 5.

Ad computandum K utimur formula:

$$\sqrt{\frac{2K}{\pi}} = 1 + 2q \qquad \text{fund. p. } 184$$

vel

$$K = \frac{\pi}{2}\{1 + 2q\}^2.$$

ad computandum E formulis:

$$\begin{aligned} \tfrac{2K}{\pi}\tfrac{2K}{\pi} - \tfrac{2K}{\pi}\cdot\tfrac{2E}{\pi} &= 8q + 16q^2 + 32q^3 \ldots, & 135 \\ \tfrac{2K}{\pi}\cdot\tfrac{2K}{\pi} &= 1 + 8q + 24q^2 + 32q^3 \ldots, & 105 \end{aligned}$$

Subtrahendo fit:

$$\frac{2K}{\pi}\cdot\frac{2E}{\pi} = 1 + 8q^2$$

et dividendo per aequationem superiorem:

$$\frac{2E}{\pi} = \frac{1 + 8q^2}{(1 + 2q)^2}.$$

Haec est altera computandi $\Sigma(l_0)$ ratio.

Est enim:

$$\Sigma(l_0) = \frac{b}{x'}E \qquad \text{Cap. VI § 3}$$

unde fit:

$$\frac{2}{\pi}\Sigma(l_0) = b\frac{1 + 8q^2}{\{1 - 2q\}^2}.$$

§ 6.

Restat, ut v computetur; quod quoniam explicitum non existit, relatas ad id functiones trigonometricas fractionis

$$\frac{iv\pi}{2K}$$

computare placet.

Proinde novam constantem A inducere volumus, definiendam aequatione

$$\sin A \cos\frac{iv\pi}{2K} = 1$$

vel:

$$\sin\frac{iv\pi}{2K} = i\cot A \qquad \operatorname{tg}\frac{iv\pi}{2K} = i\cos A$$

e quibus intelligitur, A tantummodo computari debere.

Quam ad rem

$$\frac{\Delta iv}{\sqrt{x'}} = \frac{1+2q\cos\frac{iv\pi}{K}}{1-2q\cos\frac{iv\pi}{K}} = \frac{1-2q+4q\cos^2\frac{iv\pi}{2K}}{1+2q-4q\cos^2\frac{iv\pi}{2K}}$$
$$= \frac{(1-2q)\sin^2 A+4q}{(1+2q)\sin^2 A-4q}$$

vel quum sit:

$$\Delta iv = \frac{\Delta(v,x')}{C(v,x')} = \frac{\sin B_0}{\sin l_0} = \frac{a}{b}x' = \frac{1+\varepsilon}{1-\varepsilon}x'$$
$$(1+\varepsilon)\sqrt{x'}\{(1+2q)\sin^2 A-4q\} = (1-\varepsilon)\{(1-2q)\sin^2 A+4q\}$$
$$(1+\varepsilon)\left\{(1-2q)\sin^2 A-4q\frac{1-2q}{1+2q}\right\} = (1-\varepsilon)\{(1-2q)\sin^2 A+4q\}$$
$$2\varepsilon(1-2q)\sin^2 A = 4q\left\{(1-\varepsilon)+(1+\varepsilon)\frac{1-2q}{1+2q}\right\}$$
$$\sin^2 A = \frac{2q}{\varepsilon}\cdot\frac{1}{1-4q^2}\{1-\varepsilon+2q-2q\varepsilon+1-2q+\varepsilon-2q\varepsilon\}$$
$$\sin^2 A = \frac{4q}{\varepsilon}\cdot\frac{1-2q\varepsilon}{1-4q^2}.$$

§ 7.

Praeter hanc alias deducere placet formulas ad computandum angulum A.
Est:

$$\sqrt{x'}\frac{Siv}{Civ} = \frac{\sin\frac{iv\pi}{2K}-q^2\sin\frac{3iv\pi}{2K}}{\cos\frac{iv\pi}{2K}+q^2\cos\frac{3iv\pi}{2K}}$$
$$= i\,\mathrm{tg}\,\frac{iv\pi}{2K}\,\frac{1-q^2\{2\cos\frac{iv\pi}{K}+1\}}{1+q^2\{2\cos\frac{iv\pi}{K}-1\}}$$
$$= i\,\mathrm{tg}\,\frac{iv\pi}{2K}\left\{1-4q^2\cos\frac{iv\pi}{K}\right\} = i\,\mathrm{tg}\,\frac{iv\pi}{2K}\left\{1+4q^2-8q^2\cos^2\frac{iv\pi}{K}\right\}$$
$$= -\cos A\{1-4q^2-2q\varepsilon\}.$$

vel quum sit:

$$i\frac{Siv}{Civ} = S(v,x') = \cos l_0 \qquad \text{Cap. VI § 4.}$$
$$\sqrt{x}\cos l_0 = \cos A\{1+4q^2-2q\varepsilon\}$$
$$\cos A = \sqrt{\cos l_0\cos B_0}\{1-4q^2+2q\varepsilon\}.$$

§ 8.

Ultima anguli A computandi ratio petitur ex formula:

$$Civ = 2\sqrt[4]{q}\cos\frac{iv\pi}{2K}\sqrt{\frac{x'}{x}}\,\frac{1+q^2\left(2\cos\frac{iv\pi}{K}-1\right)}{1-2q\cos\frac{iv\pi}{K}} \qquad \text{fund. p. 184.}$$

$$= \frac{1}{\sin A} \cdot \frac{1-2q}{1+q^2} \cdot \frac{1-3q^2+4q^2\cos^2\frac{iv\pi}{2K}}{1+2q-4q\cos^2\frac{iv\pi}{2K}}$$
$$= \frac{1}{\sin A} \cdot \frac{1-2q}{1+q^2} \cdot \frac{1-3q^2+q\varepsilon}{1+2q-\varepsilon\frac{1-4q^2}{1-2q\varepsilon}}$$
$$= \frac{\sqrt{x'}}{\sin A} \cdot \frac{1-2q\varepsilon}{1+q^2} \cdot \frac{1+q\varepsilon-3q^2}{1-\varepsilon}$$

vel cum sit:

$$Civ = \frac{1}{C(v,x')} = \frac{1}{\sin l_0}$$

$$\sin A = \frac{\sqrt{x'}\sin l_0}{1-\varepsilon}\{1-q\varepsilon-4q^2\} = \sqrt{\sin l_0 \sin B_0}\left\{1+\frac{\varepsilon^2}{2}-q\varepsilon-4q^2\right\}.$$

§ 9.

Jam vero alteram formulam computandae longitudinis periodi constituere placet. Est enim:

$$1-\frac{2}{\pi}\mathcal{A}(l_0) = \frac{2K}{\pi} iZ(iv+K) \qquad \text{cap. VI § 4.}$$
$$= \frac{-4iq\sin\frac{iv\pi}{K}}{1-2q\cos\frac{iv\pi}{K}} = -\frac{8iq\,\mathrm{tg}\,\frac{iv\pi}{2K}\cos^2\frac{iv\pi}{2K}}{1+2q-4q\cos^2\frac{iv\pi}{2K}}$$
$$= \frac{8q\cos A}{1+2q\sin^2 A-2q} = \frac{2\varepsilon\cos A}{\frac{1-2q\varepsilon}{1-2q}-\varepsilon} = 2\varepsilon\frac{1-2q}{1-\varepsilon}\cos A.$$

$$\mathcal{A}(l_0) = \frac{\pi}{2} - \pi\varepsilon\frac{1-2q}{1-\varepsilon}\cos A.$$

CAPUT VIII.

§ 1.

Capite sexto ostendimus, quantitates

$$B, \quad l, \quad T, \quad s, \quad \psi, \quad \varphi,$$

functionibus unius variabilis u determinari posse. Quas ut ad calculos vocemus, Θ functiones evolvendae sunt in series, progredientes ex ratione et dignitatum constantis q et cosinus atque sinus multiplorum fractionis

$$\frac{\pi u}{2K}$$

quam more analistorum „x“ significemus.

§ 2.

Quibus rationibus ad formulas Cap. VI adhibitis; obtinemus:

$$\sin l = \sin l_0 \frac{1-2q}{1+q^2} \cdot \frac{\cos x + q^2 \cos 3x}{1-2q\cos 2x},$$
$$\sin B = \sin B_0 \frac{1+2q}{1+q^2} \cdot \frac{\cos x + q^2 \cos 3x}{1+2q\cos 2x},$$
$$\frac{\cos B}{\sqrt{x'}\cos l} = \frac{1-2q\cos 2x}{1+2q\cos 2x},$$
$$\operatorname{tg} T = \cot l_0 \frac{1+q^2}{1+2q} \cdot \frac{1-2q\cos 2x}{\sin x - q^2 \sin 3x}.$$

§ 3.

Ad computandam extensionem, transformanda est aequatio

$$s = \frac{2x}{\pi}\Sigma(l_0) + \frac{b}{x'} Z\frac{2Kx}{\pi} \qquad \text{Cap. VI § 3.}$$

Adhibita formula

$$\frac{2K}{\pi} Z\frac{2Kx}{\pi} = \frac{4q\sin 2x}{1-2q\cos 2x}$$

fit:

$$s = \frac{2x}{\pi}\Sigma(l_0) + b\frac{4q}{(1-2q)^3} \cdot \frac{\sin 2x}{1-2q\cos 2x} \qquad \text{fund. p. 133.}$$

§ 4.

Ad longitudinem computandam, in formula:

$$\psi = -x\frac{2K}{\pi} iZ(iv+K) + \frac{i}{2}\log\frac{H(iv+K)}{H(iv-K)} \qquad \text{Cap. VI § 4.}$$

explicandae sunt H functiones in series, et pro $Z(iv+K)$ ejus valor ponendus est ex Cap. VI § 9. Inde fit:

$$\begin{aligned}\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A &= \frac{i}{2}\log\frac{\sin\left(\frac{iv\pi}{2K}+x\right) - q^2\sin 3\left(\frac{iv\pi}{2K}+x\right)}{\sin\left(\frac{iv\pi}{2K}-x\right) - q^2\sin 3\left(\frac{iv\pi}{2K}-x\right)} \\ &= \frac{i}{2}\log\frac{\sin\left(\frac{iv\pi}{2K}+x\right)}{\sin\left(\frac{iv\pi}{2K}-x\right)} + \frac{i}{2}\log\frac{1-q^2\left[2\cos\left(\frac{iv\pi}{K}+2x\right)+1\right]}{1-q^2\left[2\cos\left(\frac{iv\pi}{K}-2x\right)+1\right]}.\end{aligned}$$

$$= \frac{i}{2}\log\frac{1+\cot\frac{iv\pi}{2K}\operatorname{tg}x}{1-\cot\frac{iv\pi}{2K}\operatorname{tg}x} + \frac{i}{2}\log\left\{1-2q^2\left[\cos\left(\frac{iv\pi}{K}+2x\right)-\cos\left(\frac{iv\pi}{K}-2x\right)\right]\right\}$$
$$= \frac{i}{2}\log\frac{1-i\frac{\operatorname{tg}x}{\cos A}}{1+i\frac{\operatorname{tg}x}{\cos A}} + 2iq^2\sin\frac{iv\pi}{K}\sin 2x.$$

vel ponendo:

$$\cos A\cdot\operatorname{tg}o = \operatorname{tg}x$$
$$\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A = o + 4iq^2\operatorname{tg}\frac{iv\pi}{2K}\cos^2\frac{iv\pi}{2K}\sin 2x \qquad \text{C. VII § 6.}$$
$$\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A + \varepsilon q\cos A\sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg}x}{\cos A}\right).$$

§ 5.

Jam vidimus, quantitates:

$$l, \quad B, \quad T, \quad s, \quad \psi,$$

semper computari posse ex x, in quo uno definiendo tota versabitur disquisitio nostra. Cujus rei gravissima specimina ponimus haec tria:

I. Quod punctum lineae geodeticae latitudines habet l et B? Ad computandum x ex l et B adhibemus formulam:

$$\frac{\cos B}{\sqrt{x'}\cos l} = \frac{1-2q\cos 2x}{1+2q\cos 2x}.$$

Dissolvendo hanc aequationem, habita ratione „$2q\cos 2x$“, fit:

$$2q\cos 2x = \frac{\sqrt{x'}\cos l - \cos B}{\sqrt{x'}\cos l + \cos B}$$

$$\cos 2x = \frac{1+\sqrt{x'}}{1-\sqrt{x'}}\cdot\frac{\sqrt{x'}\cos l-\cos B}{\sqrt{x'}\cos l+\cos B}$$
$$= \frac{\sqrt{x'}\{\cos l-\cos B\}+\{x'\cos l-\cos B\}}{\sqrt{x'}\{\cos l-\cos B\}-\{x'\cos l-\cos B\}}.$$
$$\frac{1-\cos 2x}{2} = \frac{\cos B - x'\cos l}{\sqrt{x'}\{\cos l-\cos B\}-\{x'\cos l-\cos B\}}$$
$$\frac{1+\cos 2x}{2} = \sqrt{x'}\frac{\cos l-\cos B}{\sqrt{x'}\{\cos l-\cos B\}-\{x'\cos l-\cos B\}}$$
$$\operatorname{tg}^2 x = \frac{1}{\sqrt{x'}}\frac{\cos B - x'\cos l}{\cos l-\cos B}.$$

II. Quod punctum lineae geodeticae l_0 habet extensionem s?

Dividendo formulam:

$$s = \frac{2x}{\pi}\Sigma(l_0) + \frac{4qb}{(1-2q)^3} \cdot \frac{\sin 2x}{1 - 2q\cos 2x}$$

per $\frac{2}{\pi}\Sigma(l_0)$ fit

$$x = \frac{\pi s}{2\Sigma} - \frac{4q}{1+8q^2} \cdot \frac{\sin 2x}{1-2q\cos 2x}$$

e qua formula intelligitur, x paululum differre ab $\frac{\pi s}{2\Sigma}$, et posito:

$$x = \frac{\pi s}{2\Sigma} - 4qv \cdot \sin\frac{\pi s}{\Sigma}$$

v paululum discrepare ab 1. Hoc valore, loco x, reducto ad nostram formulam, fit:

$$v\sin\frac{\pi s}{\Sigma} = (1-8q^2)\frac{\sin\left(\frac{\pi s}{\Sigma} - 8qv\sin\frac{\pi s}{\Sigma}\right)}{1 - 2q\cos\left(\frac{\pi s}{\Sigma} - 8qv\sin\frac{\pi s}{\Sigma}\right)}$$

$$= (1-8q^2)\frac{\sin\frac{\pi s}{\Sigma}\left(1 - 32q^2v^2\sin^2\frac{\pi s}{\Sigma}\right) - \cos\frac{\pi s}{\Sigma}8qv\sin\frac{\pi s}{\Sigma}}{1 - 2q\cos\frac{\pi s}{\Sigma} + 2q\sin\frac{\pi s}{\Sigma}\,8qv\sin\frac{\pi s}{\Sigma}}$$

$$v\left\{1 - 2q\cos\frac{\pi s}{\Sigma} + 16q^2v\cdot\sin^2\frac{\pi s}{\Sigma}\right\} = \left\{1-8q^2\right\}\left\{1 - 32q^2v^2\sin^2\frac{\pi s}{\Sigma} - 8qv\cos\frac{\pi s}{\Sigma}\right\}$$

$$v = 1 - 8q^2 - 16q^2v^2\sin^2\frac{\pi s}{\Sigma} - 6qv\cos\frac{\pi s}{\Sigma}$$

vel posito:

$$v = 1 - 6\omega q$$

$$6\omega = 8q + 6v\cos\frac{\pi s}{\Sigma} + 16qv^2\sin^2\frac{\pi s}{\Sigma} = 8q + 16q\sin^2\frac{\pi s}{\Sigma} + 6\cos\frac{\pi s}{\Sigma} - 36\omega q\cos\frac{\pi s}{\Sigma}$$

et

$$6\omega = 6\cos\frac{\pi s}{\Sigma} + qr$$

$$r = 8 + 16\sin^2\frac{\pi s}{\Sigma} - 36\cos^2\frac{\pi s}{\Sigma} = 4\left(6 - 13\cos^2\frac{\pi s}{\Sigma}\right).$$

Quibus valoribus ad x reductis, fit ad extremum

$$x = \frac{\pi s}{2\Sigma} - 4q\sin\frac{\pi s}{\Sigma}\left\{1 - 6q\cos\frac{\pi s}{\Sigma} - 4q^2\left(6 - 13\cos^2\frac{\pi s}{\Sigma}\right)\right\}.$$

III. Quod punctum lineae geodeticae l_0 habet longitudinem ψ? Ad computandum x ex aequatione

$$\psi + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A + \varepsilon q\cos A\sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right)$$

posito:

$$\operatorname{tg} y\cos A = \operatorname{tg} t$$

fit:

$$\{\psi + 2\varepsilon t\cos A\} + 2\varepsilon\frac{1-2q}{1-\varepsilon}x\cos A - 2\varepsilon t\cos A + \varepsilon q\cos A\sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right);$$

et posito $\cos A\operatorname{tg}\{\psi + 2\varepsilon t\cos A\} = \operatorname{tg} y$

$$2\varepsilon\frac{1-2q}{1-\varepsilon}\cos A(x-y) + 2\varepsilon\frac{1-2q}{1-\varepsilon}y\cos A - 2\varepsilon t\cos A + \varepsilon q\cos A\sin 2x$$

$$= \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right) - \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} y}{\cos A}\right) = \operatorname{arc}\left(\operatorname{tg} = \frac{\frac{\operatorname{tg} x}{\cos A} - \frac{\operatorname{tg} y}{\cos A}}{1 + \frac{\operatorname{tg} x}{\cos A}\cdot\frac{\operatorname{tg} y}{\cos A}}\right)$$

$$= \operatorname{arc}\left(\operatorname{tg} = \cos A\frac{\operatorname{tg} x - \operatorname{tg} y}{\cos^2 A + \operatorname{tg} x\operatorname{tg} y}\right) = \operatorname{arc}\left(\operatorname{tg} = \frac{\cos A\sin(x-y)}{\cos^2 A\cos x\cos y + \sin x\sin y}\right)$$

e qua aequatione elucet, differentiam $x-y$, continentem factorem q^2, esse minimam quantitatem; quare:

$$2\varepsilon\cos A(x-y) + 2\varepsilon\frac{1-2q}{1-\varepsilon}y\cos A - 2\varepsilon t\cos A + \varepsilon q\sin 2x\cos A = \frac{\cos A(x-y)}{\cos^2 A\cos^2 y + \sin^2 y}$$

$$2\varepsilon\frac{1-2q}{1-\varepsilon}y - 2\varepsilon t + \varepsilon q\sin 2x = (x-y)\left\{\frac{1}{1-\sin^2 A\cos^2 y} - 2\varepsilon\right\}$$

$$2\frac{1-2q}{1-\varepsilon}y - 2t + q\sin 2x = (x-y)\left\{\frac{1}{\varepsilon - 4q\cos^2 y} - 2\right\}$$

$$x - y = \left\{2\frac{1-2q}{1-\varepsilon}y - 2t + q\sin 2y\right\}\frac{\varepsilon - 4q\cos^2 y}{1-2\varepsilon+8q\cos^2 y}.$$

CAPUT IX.

§ 1.

Postquam superioribus rationes, inter variabiles definitae lineae geodeticae intercedentes, propositae sunt, hoc ultimo capite elementa ipsa e quibusdam notis determinare placet.

Hic consistamus in solvenda ea, quae prima occurrit, quaestione, scilicet:

„Quae est latitudo l_0 circuli, quem tangit linea omnium brevissima intra puncta A“

„(l_1, φ_1) et B (l_2, ψ_2)?“

Quae tota quaestio eo referenda est, ut l_0 computemus e formula:

$$\psi_2 - \psi_1 = \psi = \int_{\varphi_1}^{\varphi_2} d\varphi\,\cos l_0\frac{\sqrt{1-e^2+e^2\sin^2 l_0\cos^2\varphi}}{1-\sin^2 l_0\cos^2\varphi}$$

posito:

$$\sin l_1 = \sin l_0\cos\varphi_1, \qquad \sin l_2 = \sin l_0\cos\varphi_2.$$

Quamquam e posterioribus duabus aequationibus $\cos\varphi_1$ et $\cos\varphi_2$ determinari possunt, tamen propter periodicitatem functionum trigonometricarum ipsi anguli φ_1 et φ_2 determinabiles non sunt.

Jam igitur limites definiantur necesse est et qui continent φ_1 et qui φ_2. Quamobrem praesumimus (id, quod conceditur, quum omne punctum conversionis lineae geodeticae pro ejus exordio haberi possit) primum, intra exordium et punctum A (sumto $l_2 > l_1$) nullum interjectum esse punctum conversionis, tum intra puncta A et B non esse situm exordium.

Unde sequitur, ut (cf. Cap. II § 8) φ_1 contineatur limitibus 0 et π; ut φ_2, cum $\varphi_2 - \varphi_1 < \pi$ sit (cf. Cap. V § 7) contineatur limitibus aut π et 2π, aut 0 et π, prout punctum conversionis intra puncta A et B situm est, aut non est; hoc est, prout

$$\psi > \text{ aut } < \int_{l_1}^{l_2} dl\, \frac{\cos l_0}{\cos l} \frac{\sqrt{1 - e^2\cos^2 l}}{\sqrt{\sin^2 l_0 - \sin^2 l}}$$

est, quod integrale valore suo non longe recedit a summa

$$\operatorname{arc}(\cos = \operatorname{tg} l_1 \cot l_0) - \frac{e^2}{2}\cos l_0 \operatorname{arc}\left(\cos = \frac{\sin l_1}{\sin l_0}\right).$$

Sed ut revertamur, unde defleximus, l_0 computemus. Posito:

$$\chi = \int_{\varphi_1}^{\varphi_2} d\varphi\, \cos l_0 \frac{1 - \sqrt{1 - e^2 + e^2\sin^2 l_0 \cos^2\varphi}}{1 - \sin^2 l_0 \cos^2\varphi}$$

aequatio nostra transit in:

$$\psi + \chi = \int_{\varphi_1}^{\varphi_2} \frac{\cos l_0\, d\varphi}{1 - \sin^2 l_0 \cos^2\varphi} = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg}\varphi_2}{\cos l_0}\right) - \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg}\varphi_1}{\cos l_0}\right)$$

vel, cum sit:

$$\operatorname{arc}(\operatorname{tg} = x) = \operatorname{arc}\left(\cos = \frac{1}{\sqrt{1 + x^2}}\right)$$

$$\psi + \chi = \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_2 + \cos^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_1 + \cos^2 l_0}}\right)$$

vel, cum sit:

$$\operatorname{arc}(\operatorname{tg} = x) = \operatorname{arc}\left(\cos = \frac{1}{\sqrt{1 + x^2}}\right)$$

$$\psi + \chi = \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_2 + \cos^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\operatorname{tg}^2\varphi_1 + \cos^2 l_0}}\right)$$

$$= \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{1}{\cos^2\varphi_2} - \sin^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{1}{\cos^2\varphi_1} - \sin^2 l_0}}\right)$$

$$= \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{\sin^2 l_0}{\sin^2 l_2} - \sin^2 l_0}}\right) - \operatorname{arc}\left(\cos = \frac{\cos l_0}{\sqrt{\frac{\sin^2 l_0}{\sin^2 l_1} - \sin^2 l_0}}\right)$$

$$= \operatorname{arc}(\cos = \cot l_0 \operatorname{tg} l_2) - \operatorname{arc}(\cos = \cot l_0 \operatorname{tg} l_1)$$

quam aequationem ad elegantiorem formam reducere possumus usi notissima illa identitate:

$$\begin{vmatrix} 1 & \cos x & \cos y \\ \cos x & 1 & \cos(x-y) \\ \cos y & \cos(x-y) & 1 \end{vmatrix}$$

in qua posito:

$$x = \cot l_0 \operatorname{tg} l_2, \qquad y = \cot l_0 \operatorname{tg} l_1, \qquad x - y = \psi + \chi$$

fit:

$$\begin{vmatrix} 1 & \operatorname{tg} l_2 \cot l_0 & \operatorname{tg} l_1 \cot l_0 \\ \operatorname{tg} l_2 \cot l_0 & 1 & \cos(\psi+\chi) \\ \operatorname{tg} l_1 \cot l_0 & \cos(\psi+\chi) & 1 \end{vmatrix} = \begin{vmatrix} \operatorname{tg}^2 l_0 & \operatorname{tg} l_2 & \operatorname{tg} l_1 \\ \operatorname{tg} l_2 & 1 & \cos(\psi+\chi) \\ \operatorname{tg} l_1 & \cos(\psi+\chi) & 1 \end{vmatrix} = 0$$

vel:

$$\text{I.} \qquad \operatorname{tg}^2 l_0 \sin^2(\psi+\chi) = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos(\psi+\chi).$$

Qua in aequatione l_0 occurrit et in „$\operatorname{tg}^2 l_0$“ et in „χ“. Sin hoc temporis momento non stricte libet definire l_0, sed valorem quendam l_0', qui ipsi propinquus est; χ in formula negligi potest, quod continens factorem e^2 minimum est.

Ergo l_0', definitum e formula:

$$\text{II.} \qquad \operatorname{tg}^2 l_0' \sin^2 \psi = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos \psi$$

differt a l_0 quantitate minima secundi gradus, si comparas cum excentricitate.

Ad computandum l_0'', propius accedens valore suo ad l_0, non plane abjiciendum est χ, sed habenda ratio est membri

$$\frac{e^2}{2} \cos l_0(\varphi_2 - \varphi_1)$$

seriei, secundum dignitates quadrati e^2 progredientis, in quam integrale χ explicari potest. Tum valor l_0'', ex aequatione hac definitus:

$$\text{III.} \quad \operatorname{tg}^2 l_0'' \sin^2\left[\psi + \frac{e^2}{2} \cos l_0''(\varphi_2 - \varphi_1)\right] = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos\left[\psi + \frac{e^2}{2} \cos l_0''(\varphi_2 - \varphi_1)\right]$$

a l_0 recedit quantitate, quae, comparata cum excentricitate, minima quarti gradus est.

Primo obtutu putes graves difficultates obstare computationi quantitatis l_0'' ex aequatione III. Si quidem l_0'' non modo in $\operatorname{tg}^2 l_0$ sed etiam in:

$$\frac{e^2}{2} \cos l_0''(\varphi_2 - \varphi_1)$$

inest et altero quidem loco tum in $\cos l_0''$ tum in angulis φ_2 et φ_1, quos formulis:

$$\sin l_0'' \cos\varphi_2 = \sin l_2, \qquad \sin l_0'' \cos\varphi_1 = \sin l_1$$

definiri notum est. Attamen, quod

$$\frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)$$

minimum est, ea, quae hoc continentur, computari licet sine vitio, loco l_0'', jam nota quantitate l_0' posita.

Fieri potest, ut l_0' definitum ex aequatione II valorem, prope l_0 situm, non exhibeat, tum quidem, cum summa $l_1 + l_2$ minima est et ψ non longe a π differt.

Ubi $\operatorname{tg}^2 l_0$ repraesentatur, ut facile intelligi potest, quotientem quantitatum minimarum, unde sequitur, ut vel minimo incremento, addendo integrale χ ad quantitatem ψ, adducto, magna valoris l_0 mutatio existat.

Quodsi accidit, l_0 computandi causa ad ipsam formulam II confugiendum est; quae quia, ponendo loco l_0'' valorem prope accedentem l_0' jamque notum, ad elegantiorem formam redigi non potest, alia utendum est ratione scilicet ponendi pro quibusdam quantitatibus alias his proxime sitas.

Quoniam quantitatem

$$\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)$$

in aequatione III, non multo differentem a π, possumus, sumere licet:

$$\sin\left[\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right] = \pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)$$

$$\cos\left[\psi + \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right] = -1 + \frac{1}{2}\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2$$

unde fit:

$$\operatorname{tg}^2 l_0''\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2 = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 + 2\operatorname{tg} l_1 \operatorname{tg} l_2 - \operatorname{tg} l_1 \operatorname{tg} l_2\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2$$

$$\left\{\operatorname{tg}^2 l_0'' + \operatorname{tg} l_1 \operatorname{tg} l_2\right\}\left\{\pi - \psi - \frac{e^2}{2}\cos l_0''(\varphi_2 - \varphi_1)\right\}^2 = \left\{\operatorname{tg} l_1 + \operatorname{tg} l_2\right\}^2$$

Atqui l_0 non multum differt a $\pi - l_0''$; ergo summa $\cos\varphi_1 + \cos\varphi_2$ minima est, quare aut $\varphi_2 + \varphi_1$ aut $\varphi_2 - \varphi_1$ non longe recedit a π. Quod cum in summam cadere nequeat, siquidem simplex angulus φ_2 major est quam π (quippe longitudinem ψ posuimus prope π jacentem, itaque.

$$\psi > \int_{l_1}^{l_2} dl\,\frac{\cos l_0}{\cos l}\,\frac{\sqrt{1 - e^2\cos^2 l}}{\sqrt{\sin^2 l_0 - \sin^2 l}} \qquad \text{cf. p. 35}$$

esse) differentia „$\varphi_2 - \varphi_1$“ prope π sita est, ita ut loco $\varphi_2 - \varphi_1$ ponere liceat π.

Unde fit:

$$\left\{\operatorname{tg}^2 l_0'' + \operatorname{tg} l_1 \operatorname{tg} l_2\right\}\left\{\pi - \psi - e^2\frac{\pi}{2}\cos l_0''\right\}^2 = \{\operatorname{tg} l_1 + \operatorname{tg} l_2\}^2$$

$$\cos^2 l_1 \cos^2 l_2 \left\{\frac{1}{\cos^2 l_0''} - \frac{\cos(l_1+l_2)}{\cos l_1 \cos l_2}\right\}\left\{\pi - \psi - \frac{e^2}{2}\pi\cos l_0''\right\}^2 = \sin^2(l_1+l_2)$$

$$\cos l_1 \cos l_2 \left\{\frac{\cos l_1 \cos l_2}{\cos^2 l_0''} - \cos(l_1+l_2)\right\}\left\{\pi - \psi - \frac{e^2\pi}{2}\cos l_0''\right\}^2 = \sin^2(l_1+l_2)$$

vel posito:

$$\text{loco } \cos l_2 : \cos l_1; \quad \text{loco } \cos(l_1+l_2) : 1; \quad \text{loco } \sin(l_1+l_2) : l_1+l_2,$$

$$\cos^2 l_1 \left\{\frac{\cos^2 l_1}{\cos^2 l_0''} - 1\right\}\left\{\pi - \psi - e^2\frac{\pi}{2}\cos l_0''\right\}^2 = (l_1+l_2)^2$$

$$\cos l_1 \cot T \left\{\pi - \psi - \frac{e^2}{2}\pi\cos l_0''\right\} = l_1 + l_2.$$

Qua in formula T constat significare Azimut geodeticae lineae l_0'' in puncto A; ergo spectat in l_0'' formula:

$$\sin T \cos l_1 = \cos l_0''.$$

Posito hoc valore quantitatis $\cos l_0''$ in nostra aequatione, existit formula:

$$\cos l_1 \cot T\{\pi - \psi\} - e^2\frac{\pi}{2}\cos^2 l_1 \cos T = l_1 + l_2$$

e qua T itemque l_0'' facile computari possunt.

Illa aequatio petitur ex his rationibus geometricis. — Linea geodetica l_2, producta ultra punctum B, donec circulum „$-l_0$“ (e. g. in C) secet, cum circulo $-l_1$ et cum meridiano puncti B (quas curvas in D secantes facimus) efficit rectangulum triangulum, cujus latera tam brevia sunt, ut planus, possit triangulus intelligi et sumi:

$$\frac{BD}{CD} = \operatorname{tg} BCD \qquad \text{cf. cap. I § 9.}$$

Atqui, ut facile intelligitur, est:

$$BD = a\int_{-l_1}^{l_2} dl\sqrt{1 - e^2\cos^2 l}$$

$$CD = a\cos l_1\{2\mathcal{A}(l_0) - \psi\}$$

et angulus BCD complimentum ipsius T efficit. Quibus relatis in nostram formulam obtinemus:

$$\cot T \cos l_1\{2\mathcal{A}(l_0) - \psi\} = \int_{-l_1}^{l_2} dl\sqrt{1 - e^2\cos^2 l}$$

vel neglecta excentricitate sub signo integrationis et in serie longitudinis periodi evoluta cap. III. tantummodo priorum duorum membrorum habita ratione,

$$\cot T \cos l_1\left\{\pi - e^2\frac{\pi}{2}\cos l_0 - \psi\right\} = l_1 + l_2,$$

id, quod erat demonstrandum.

Postquam aliqua ratione valor l_0'' computatus est, qui differt a l_1, quantitate minima, ratione habita excentricitatis quarti gradus, latitudinem l_0 facile est accuratius computatu.

Si, evolventes longitudinem e theoremate Tayloriano, secundum dignitates differentiae l_0'', post secundum membrum seriem interrumpimus, obtinemus formulam:

$$\psi = \int_{\varphi_1}^{\varphi_2} d\varphi \cos l_0'' \frac{\sqrt{1 - e^2 + e^2 \sin^2 l_0'' \cos^2 \varphi}}{1 - \sin^2 l_0'' \cos^2 \varphi} + (l_0 - l_0'') \frac{\partial \psi}{\partial l_0}$$

vel ponendo et pro integrali et pro $\frac{\partial \psi}{\partial l_0}$ valores:

$$\text{V.} \qquad l_0 = l_0'' + \frac{t_2 - t_1 - \psi}{u}$$

qua in formula:

VI.	$t_2 + 2\varepsilon \frac{1-2q}{1-\varepsilon} x_2 \cos A + \varepsilon q \sin 2x_2 \cos A$	$=$	$\operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x_2}{\cos A}\right)$	cf. cap. VIII § 4
VII.	$t_1 + 2\varepsilon \frac{1-2q}{1-\varepsilon} x_1 \cos A + \varepsilon q \sin 2x_1 \cos A$	$=$	$\operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x_1}{\cos A}\right)$	
VIII.	$u \sin B_0$	$=$	$8q(x_2 - x_1) + (1 - 4q)\{\cot x_2 - \cot x_1\}$	cf. cap. VIII § 3.

Jam vero, ut vim formularum exemplo illustremus, sit:

$$B_1 = 48^\circ 43' 5'', \qquad B_2 = 57^\circ 4' 5'', \qquad \psi = 79^\circ 5' 20''$$

quibus praemissis fit: e formula:

$$a \operatorname{tg} l = b \operatorname{tg} B \quad \text{cap. I § 7}$$

$$l_1 = 48^\circ 37' 22{,}505'' \qquad l_2 = 56^\circ 58' 49{,}647''$$

" " $\quad \operatorname{tg}^2 l_0' \sin^2 \psi = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos \psi \quad$ cap. IX f. II

$$l_0' = 60^\circ 25' 47''$$

" " $\quad \sin l = \sin l_0 \cos \varphi \quad$ cap. II § 7

$$\varphi_1 = 149^\circ 24' 10'' \qquad \varphi_2 = 195^\circ 25' 33''$$

$$\psi + \frac{e^2}{2} \cos l_0' (\varphi_2 - \varphi_1) = 79^\circ 10' 10''$$

$$\operatorname{tg}^2 l_0'' \sin^2 \left[\psi + \frac{e^2}{2} \cos l_0' (\varphi_2 - \varphi_1)\right] = \operatorname{tg}^2 l_1 + \operatorname{tg}^2 l_2 - 2 \operatorname{tg} l_1 \operatorname{tg} l_2 \cos \left[\psi + \frac{e^2}{2} \cos l_0' (\varphi_2 - \varphi_1)\right]$$

$$l_0'' = 60^\circ 26' 34'' \qquad \text{cap. IX f. III}$$

e formula:

$$a \operatorname{tg} l = b \operatorname{tg} B \qquad \text{cap. I § 7.}$$
$$B_0'' = 60^\circ 31' 31{,}1''$$

" " $\quad x' \cos l_0 = \cos B_0 \qquad$ cap. II § 9.
$$x' = 0{,}9974675491$$

" " $\quad 2q = \frac{1-\sqrt{x'}}{1+\sqrt{x'}} \qquad$ cap. VII § 4.
$$q = 0{,}0003169603$$

" " $\quad \cos A = \sqrt{\cos l_0 \cos B_0}\{1 + 2q\varepsilon - 4q^2\} \qquad$ cap. VII § 7.
$$\log \cos A = 9{,}6925530966$$

" " $\quad \sqrt{x'} \operatorname{tg}^2 x = \frac{\cos B - x' \cos l}{\cos l - \cos B} \qquad$ cap. VIII, § 5, I.

$x_1 = 149^\circ 38' 429{,}65''; \qquad x_2 = 195^\circ 25' 10{,}297''$

$$t + 2\varepsilon \frac{1-2q}{1-\varepsilon} x \cos A + \varepsilon q \cos A \sin 2x = \operatorname{arc}\left(\operatorname{tg} = \frac{\operatorname{tg} x}{\cos A}\right) \qquad \text{cap. IX, f. VI et VII.}$$

$t_1 = 130^\circ 10' 28{,}161''; \qquad t_2 = 209^\circ 16' 47{,}599''; \qquad t_2 - t_1 - \psi = 59{,}438''$

$$u \sin B_0 = 8q(x_2 - x_1) + (1 - 4q)(\cot x_2 - \cot x_1) \qquad \text{cap. IX f. VIII.}$$
$$u = 6{,}125$$
$$l_0 = l_0'' + \frac{t_2 - t_1 - \psi}{u} \qquad \text{cap. IX f. V.}$$
$$l_0 = 60^\circ 26' 43{,}97''.$$

V I T A.

Natus sum, Paulus Albertus Gordan, fidei evangelicae addictus, Vratislaviae MDCCCXXXVII mens. April. die XXVII, patre David mercatore, matre Friderica e gente Friedenthal, quam mihi jam puero morte praematura ereptam valde lugeo; patre adhuc vigente maxime gaudeo.

Litterarum elementa a praeceptoribus singulis usque ad duodecimum annum doctus, tum Gymnasium Fridericianum Vratislaviae, Vimmero rectore, qui ibi adhuc floret, frequentavi. Duobus annis post, quum, quarta et partim tertia classi absolutis, a patre permotus, ut mercaturae me traderem, tum Berolini, tum Vratislaviae in patris negotiis res mercatori necessarias discerem, litteris tamen semper amore quodam vacabam. Illo tempore mercaturae quoque scholae disciplinis affui. Quem cursum quum absolvissem, et ut lingua Gallica melius perdiscerem uti, et ut terras exsternas cognoscerem a patre Genevam missus sum, ubi duos annos morabar.

Geneva ad patris negotia reversus, quum a pueris maximis cum deliciis amplecterer mathematicam, hanc voluntatem sequens, privatim a professore Schellbach in hac disciplina sum eruditus. A quo viro excellentissimo commotus, ut totum me traderem studiis mathematicis, patris negotia reliqui, atque universitatem Berolinensem, testimonium maturitatis nondum adeptus, frequentare statui. Ubi cum per tria semestria professorem Kummer diligenter audivissem, me contuli Neissam, ubi semestre in prima gymnasii classi versatus maturitatis testimonio instructus sum.

Quatuor per annos ab illo tempore peractos vocabam studio mathematico; primum Vratislaviae, tum Regiomonti, deinde rursus Vratislaviae his affui lectionibus.

Mathematicis:	Ill. Kummer, Joachimsthal, Richelot, Rosenhain, Schroeter, Galle.
Physicis:	Ill. Frankenhein, Neumann.
Astronomicis:	Ill. Galle.
Mineralogicis:	Ill. Roemer.
Philosophicis:	Ill. Braniss.
Politicis:	Ill. Tellkampf.
Historicis:	Ill. Junkmann.

Quibus omnibus de me optime meritis viris gratias maximas habeo atque semper habebo.

In certamine litterarum ab amplissimo philosophorum ordine almae Viadrinae:

• De linea geodetica •

instituto in arenam descendi atque opusculum hac de quaestione a me compositum a. d. III Non. Aug. anni praeteriti palma ornatum est.

THESES. •

I. Functionum divisio, ab Ill. Cournot in empiricas et analyticas proposita, non apta videatur esse, quia functiones empiricae omnino non dantur.

II. Methodum infinite parvorum non minus accuratam esse contendo, quam methodum limitum.

III. Magis interest, quaerere, quae qualitates insint functioni, differentiali aequatione definitae; quam quibus functionibus jam notis exprimi possit.

IV. Principia Democriti usque ad nostra tempora sunt fundamenta doctrinae atomorum.